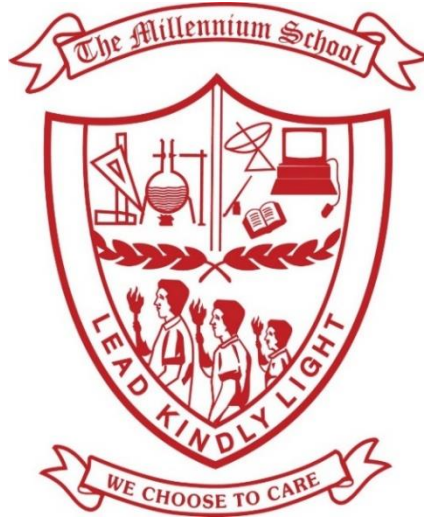




مدرسة الألفية
The Millennium School
A GEMS SCHOOL



COMPUTER SCIENCE
CBSE BOARD PROJECT FILE
2024-2025



Name: _____

Grade: _____

Roll no: _____

Acknowledgement

I take this opportunity to sincerely express my gratitude to all who have helped me complete this project .

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COMPUTER SCIENCE

CERTIFICATE

Certified to be the bonafide record of the project work done by Master / Miss _____ of Grade XII as prescribed by the Central Board of Secondary Education, Delhi during the academic year 2024- 2025.

Teacher in-charge

Principal

Submitted for the Practical Examination on _____
held in **The Millennium School, Dubai**

Internal Examiner

External Examiner

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Arun Jose, Aditya Menon, Kavin Senthilkumar

Title of the Project

Black Mamba Arcade

Aim of the Project

"Black Mamba Arcade, created by Akagames, is an innovative app designed to help students manage the stress and anxiety that often accompany exam season. Recognizing the intense pressures students face, Black Mamba Arcade offers an array of engaging and immersive arcade games aimed at reducing stress through interactive entertainment. These games provide a mental escape, allowing students to unwind, recharge, and regain focus while tackling their academic challenges.

Beyond offering an enjoyable gaming experience, Black Mamba Arcade incorporates a unique educational element through its integrated merchandise shop. This feature enables students to explore and practice real-world money management in a simulated, risk-free environment. Players earn and spend virtual currency, make purchasing decisions, and learn to balance rewards and savings, reinforcing financial literacy in a way that's both practical and enjoyable. By promoting these essential life skills, Black Mamba Arcade empowers students to build confidence not only in managing finances but also in approaching real-life challenges with a balanced, thoughtful mindset.

Through its dual focus on stress relief and financial education, Black Mamba Arcade fosters emotional resilience, supports academic productivity, and encourages healthy habits, equipping students with tools to thrive both academically and personally."

System Requirements

Hardware:

- Laptop/computer with Windows 10 OS
- 64-bit CPU

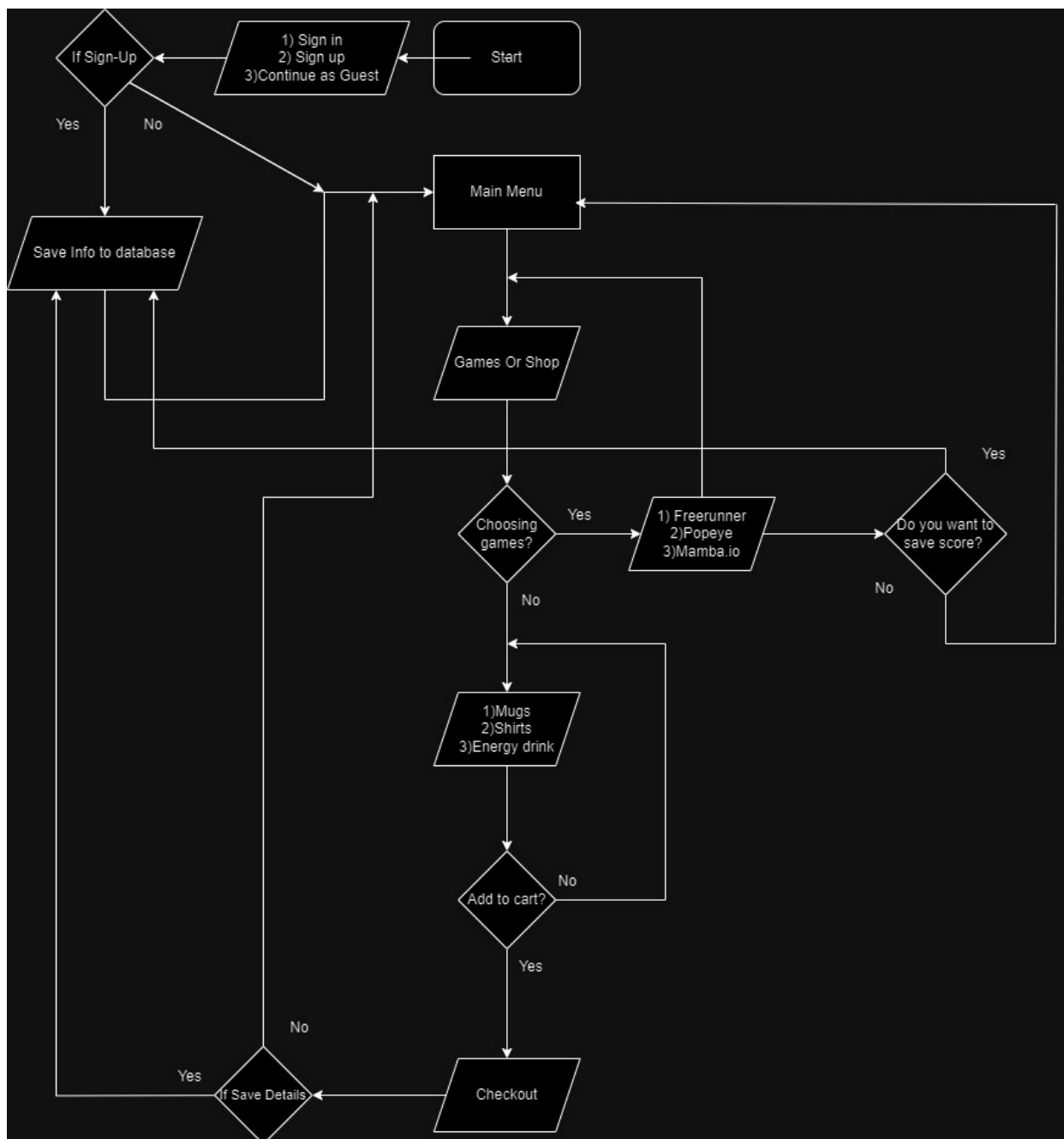
Software:

- Python programming language
- Pygame module (for game development)
- MySQL 8.4 or newer(UNICODE COMMAND LINE)
- MySQL connectivity module (for database management)

Installation Procedure

- **Install Python:** Download and install Python from python.org.
- **Install Pygame:** Use the command pip install pygame in the command prompt or terminal to install Pygame.
- **MySQL Setup:** Ensure MySQL is installed on the system, then install MySQL connectivity module using pip install mysql-connector-python.
- **Download the Project Files:** Obtain the source files for Black Mamba Arcade and place them in a working directory.
- **Run the Program:** Run the python program named BLACKMAMBA.py and enjoy :)

System Layout



Functions and Modules Used:

Python Functions & Modules:

- **Random Module**
- **MySQL connector module**
- **Pygame Module**
- **Math Module**

Game Functions:

- **games_main():** Manages the main games menu.
- **controls_poppeye():** Loads and displays controls for the "Poppeye" game.
- **controls_george():** Loads and displays controls for the "George" game.
- **controls_mambaio():** Loads and displays controls for the "Mambaio" game.
- **poppeye_game():** Main function for the "Poppeye" game.
- **george_game():** Main function for the "George" game.
- **mambaio_game():** Main function for the "Mambaio" game.
- **game_over():** Handles the end game screen.

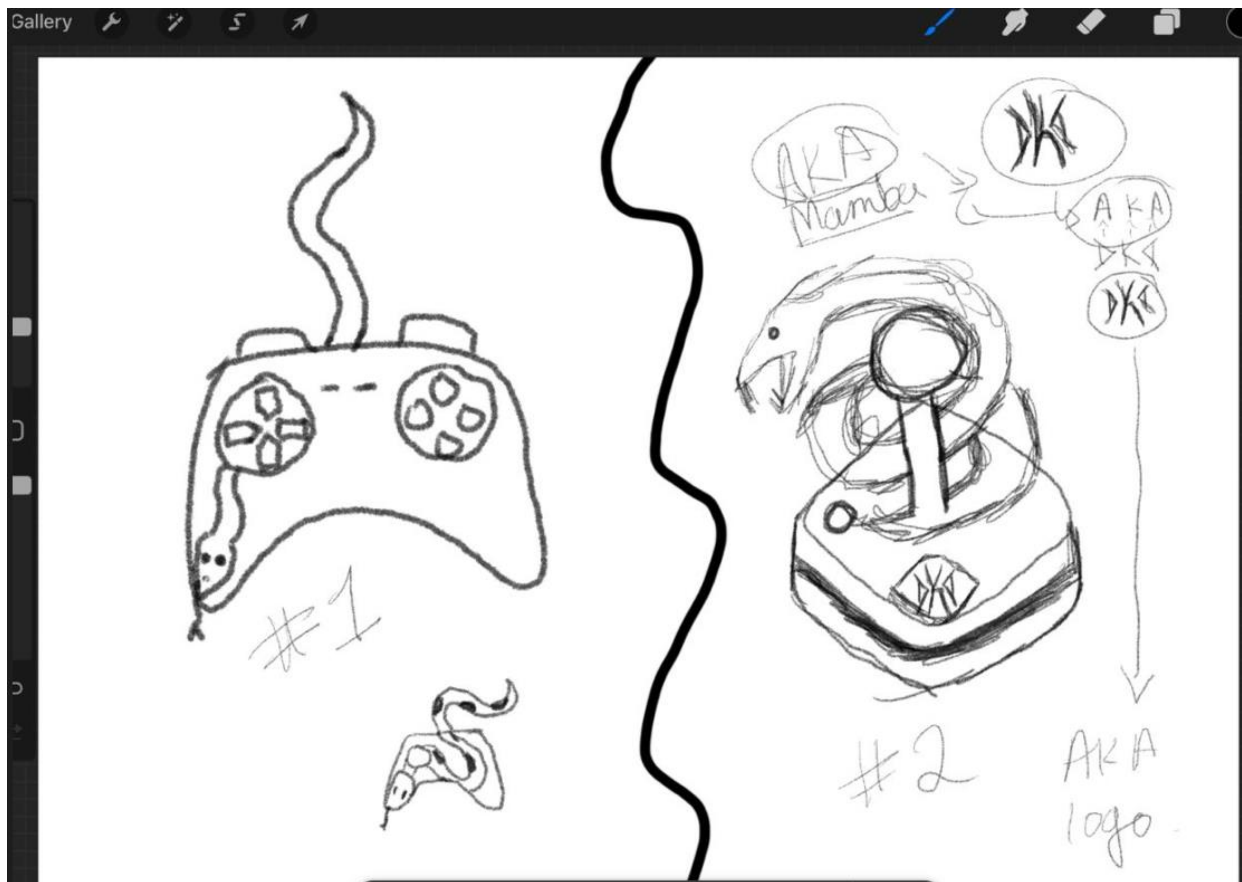
UI Functions:

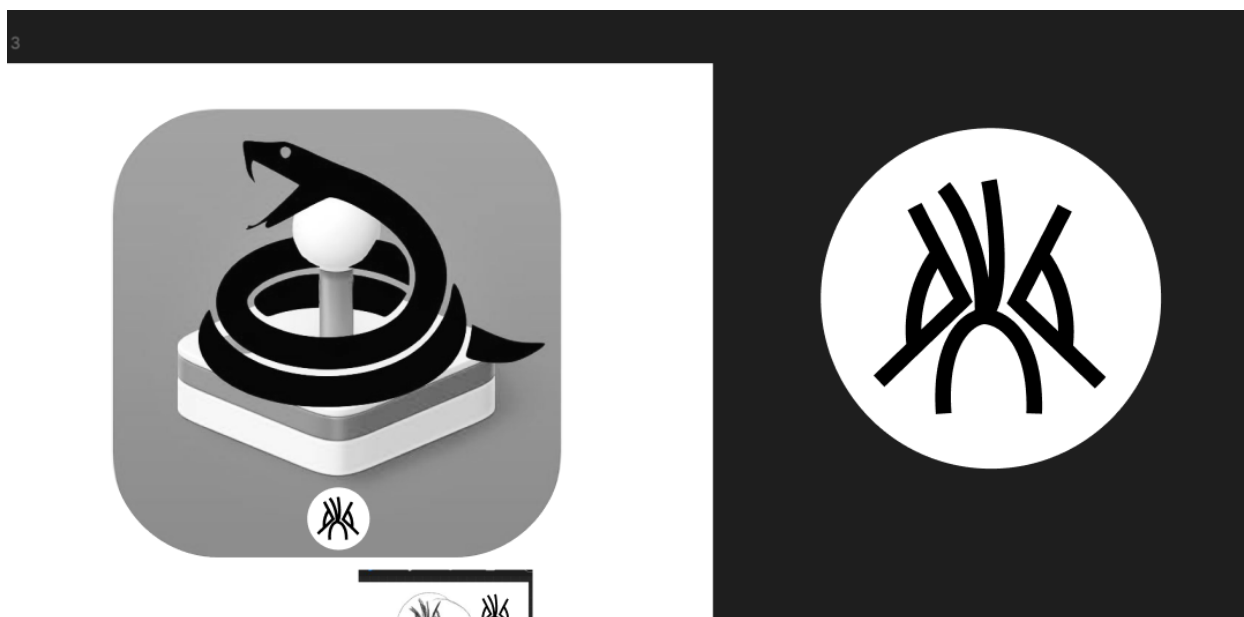
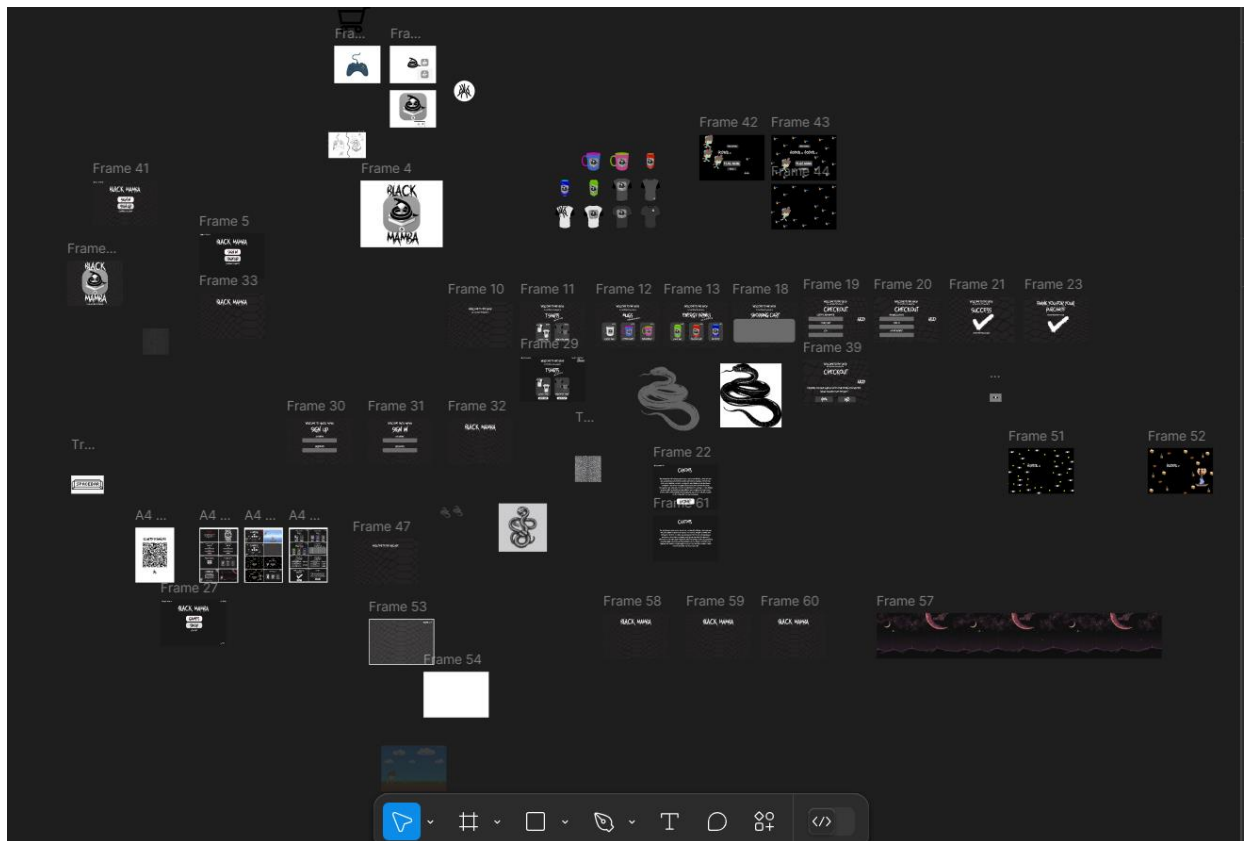
- **shop_main():** Displays the main shop screen with categories.
- **shop_shirt():** Displays the shirt shop UI.
- **shop_mugs():** Displays the mug shop UI.
- **checkout_bill():** Shows the checkout bill UI.
- **checkout_address():** Allows input of address details in checkout.
- **checkout_credit():** Allows input of credit card details in checkout.
- **emptybill():** Clears the bill and displays a cleared checkout screen.
- **saved_checkout():** Displays the saved checkout screen.
- **warning():** Shows a warning screen for epilepsy
- **startup():** Shows the initial startup screen.
- **mamba_signin():** Provides the user sign-in screen.
- **mamba_signup():** Provides the user sign-up screen.

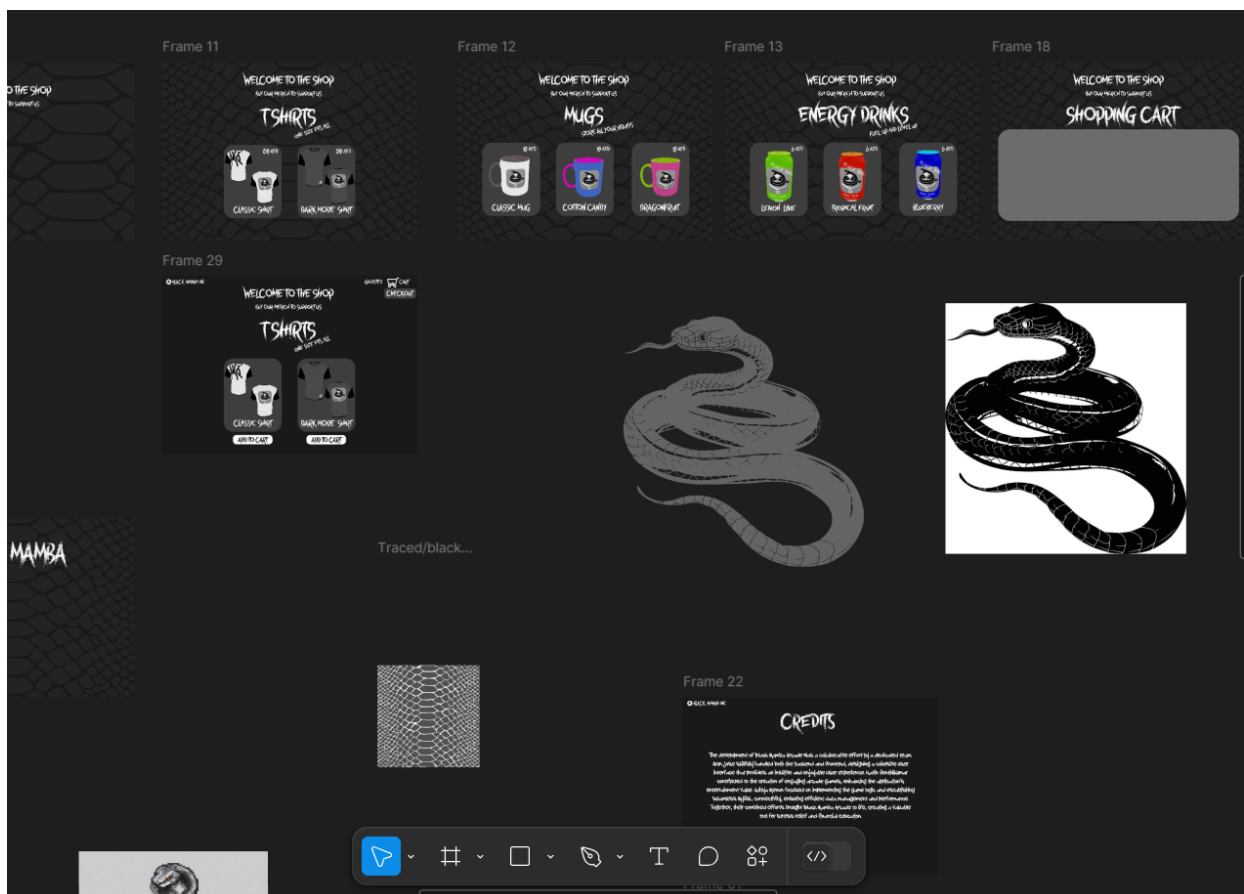
Database Functions:

- **checkout_bill():** Saves and loads checkout data from the database.
- **saved_checkout():** Updates and retrieves saved checkout details.
- **checkout_address():** Updates and retrieves user address details.
- **mamba_signin():** Authenticates user credentials against the database.
- **mamba_signup():** Registers new users in the database.

Initial stages of the Project Design







Script:

```
1. #HEHE WELCOME TO BLACK MAMBA
2. import pygame,sys
3. import mysql.connector
4. import random as ran
5. import math
6. mysql = mysql.connector.connect(host = 'localhost', user = 'root', passwd =
'tiger',database = 'blackmamba')
7. cursor = mysql.cursor()
8. pygame.init()
9. screen = pygame.display.set_mode((1000,700))
10. clock = pygame.time.Clock()
11. main_font = pygame.font.Font('fonts\\nightcore.ttf',150)
12. ICON = pygame.image.load('graphics\\AKAGAMES.png')
13. sub_font = pygame.font.Font('fonts\\nightcore.ttf',22)
14. guest_font = pygame.font.Font('fonts\\nightcore.ttf',32)
15. bill_font = pygame.font.Font('fonts\\adripl.ttf',33)
16. text_input_font = pygame.font.Font('fonts\\adripl.ttf',66)
17. total_font = pygame.font.Font('fonts\\adripl.ttf',56)
18. number_font = pygame.font.Font('fonts\\adripl.TTF',16)
19. finalprice_font = pygame.font.Font('fonts\\adripl.ttf',92)
20. global username, cartlist, cartno, merchandise, total,
password,username_input,password_input
21. username = "guest"
22. cartno = 0
23. credentials = []
24. cartlist = {101:0,102:0,201:0,202:0,203:0,301:0,302:0,303:0}
25. merchandise = {101:["Classic Shirt",30], 102:["Dark Mode Shirt",30],
201:["Classic Mug",10], 202: ["Cotton Candy Mug",10], 203:["Dragon Fruit Mug",10],
301: ["Lemon Lime Energy Drink",6], 302 : ["Tropical Fruit Energy Drink",6], 303:
["BlueBerry Energy Drink",6]}
26.
27. def mambacredits():
28.     global username, cartlist, cartno, merchandise
29.
30.     pygame.display.set_icon(ICON)
31.     pygame.display.set_caption("CHECKOUT")
32.     background = pygame.image.load('graphics\\credits.png').convert_alpha()
33.
34.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
35.     home_rect = home.get_rect(topleft = (17,8))
36.
37.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
38.     homehov_rect = homehov.get_rect(topleft = (11,8))
39.
40.     home_big = pygame.image.load('graphics\\home.png').convert_alpha()
41.     home_big_rect = home_big.get_rect(topleft = (368,565))
42.
43.     username_disp = sub_font.render(username,True, "White")
44.     username_rect = username_disp.get_rect(topright = (872,7))
45.
46.
47.     while True:
48.
49.         screen.blit(background,(0,0))
50.         screen.blit(home,home_rect)
51.         screen.blit(username_disp,username_rect)
52.         screen.blit(home_big,home_big_rect)
53.
54.         for event in pygame.event.get():
55.             if event.type == pygame.QUIT:
56.                 pygame.quit()
57.                 sys.exit()
58.             if event.type ==pygame.MOUSEBUTTONDOWN:
59.                 if home_rect.collidepoint(event.pos):
60.                     mamba_main
```

```

61.         if event.type ==pygame.MOUSEBUTTONDOWN:
62.             if home_big_rect.collidepoint(event.pos):
63.                 mamba_main()
64.
65.
66.         pygame.display.update()
67.         clock.tick(60)
68.
69. def games_main():
70.     clock = pygame.time.Clock()
71.     pygame.display.set_icon(ICON)
72.     pygame.display.set_caption("MAMBA GAMES")
73.     background = pygame.image.load('graphics\\arcade.png').convert_alpha()
74.
75.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
76.     home_rect = home.get_rect(topleft = (17,8))
77.
78.
79.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
80.     homehov_rect = homehov.get_rect(topleft = (11,8))
81.
82.     george = pygame.image.load('graphics\\george.png').convert_alpha()
83.     george_rect = george.get_rect(topleft = (112,263))
84.
85.     poppeye = pygame.image.load('graphics\\poppeye.png').convert_alpha()
86.     poppeye_rect = poppeye.get_rect(topleft = (384,263))
87.
88.     mambaio = pygame.image.load('graphics\\mambaio.png').convert_alpha()
89.     mambaio_rect = mambaio.get_rect(topleft = (662,263))
90.
91.     username_disp = sub_font.render(username,True, "White")
92.     username_rect = username_disp.get_rect(topright = (948,8))
93.
94.
95.
96.     while True:
97.
98.         screen.blit(background,(0,0))
99.         screen.blit(home,home_rect)
100.        screen.blit(username_disp,username_rect)
101.        screen.blit(george,george_rect)
102.        screen.blit(poppeye,poppeye_rect)
103.        screen.blit(mambaio,mambaio_rect)
104.
105.        for event in pygame.event.get():
106.            if event.type == pygame.QUIT:
107.                pygame.quit()
108.                sys.exit()
109.
110.            if event.type ==pygame.MOUSEBUTTONDOWN:
111.                if home_rect.collidepoint(event.pos):
112.                    mamba_main()
113.
114.            if event.type ==pygame.MOUSEBUTTONDOWN:
115.                if george_rect.collidepoint(event.pos):
116.                    controls_george()
117.
118.            if event.type ==pygame.MOUSEBUTTONDOWN:
119.                if poppeye_rect.collidepoint(event.pos):
120.                    controls_poppeye()
121.
122.            if event.type ==pygame.MOUSEBUTTONDOWN:
123.                if mambaio_rect.collidepoint(event.pos):
124.                    controls_mambaio()
125.
126.        pygame.display.update()
127.        clock.tick(60)
128.

```

```

129. def controls_poppeye():
130.     clock = pygame.time.Clock()
131.     pygame.display.set_icon(ICON)
132.     pygame.display.set_caption("MAMBA GAMES")
133.     background =
pygame.image.load('graphics\\controls_poppeye.png').convert_alpha()
134.
135.
136.     while True:
137.
138.         screen.blit(background, (0,0))
139.         for event in pygame.event.get():
140.             if event.type == pygame.QUIT:
141.                 pygame.quit()
142.                 sys.exit()
143.
144.             if event.type ==pygame.KEYDOWN:
145.                 poppeye_game()
146.
147.         pygame.display.update()
148.         clock.tick(60)
149.
150. def controls_george():
151.     clock = pygame.time.Clock()
152.     pygame.display.set_icon(ICON)
153.     pygame.display.set_caption("GOERGE CONTROLS")
154.     background =
pygame.image.load('graphics\\controls_george.png').convert_alpha()
155.
156.
157.     while True:
158.
159.         screen.blit(background, (0,0))
160.         for event in pygame.event.get():
161.             if event.type == pygame.QUIT:
162.                 pygame.quit()
163.                 sys.exit()
164.
165.             if event.type ==pygame.KEYDOWN:
166.                 mambarunner()
167.
168.         pygame.display.update()
169.         clock.tick(60)
170.
171. def controls_mambaio():
172.     clock = pygame.time.Clock()
173.     pygame.display.set_icon(ICON)
174.     pygame.display.set_caption("MAMBA GAMES")
175.     background =
pygame.image.load('graphics\\controls_mambaio.png').convert_alpha()
176.
177.
178.     while True:
179.
180.         screen.blit(background, (0,0))
181.         for event in pygame.event.get():
182.             if event.type == pygame.QUIT:
183.                 pygame.quit()
184.                 sys.exit()
185.
186.             if event.type ==pygame.KEYDOWN:
187.                 mambaio_game()
188.
189.         pygame.display.update()
190.         clock.tick(60)
191.
192.
193. def mamba_main():

```

```

194.     global username, cartlist, cartno, merchandise, total, password
195.
196.
197.
198.     clock = pygame.time.Clock()
199.     pygame.display.set_icon(ICON)
200.     pygame.display.set_caption("MAMBA HOME")
201.     background = pygame.image.load('graphics\\mamba_login.png').convert_alpha()
202.
203.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
204.     home_rect = home.get_rect(topleft = (17,8))
205.
206.
207.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
208.     homehov_rect = homehov.get_rect(topleft = (11,8))
209.
210.     games = pygame.image.load('graphics\\games.png').convert_alpha()
211.     games_rect = games.get_rect(topleft = (387,238))
212.
213.     shop = pygame.image.load('graphics\\shop.png').convert_alpha()
214.     shop_rect = shop.get_rect(topleft = (387,349))
215.
216.     logout = pygame.image.load('graphics\\log_out.png').convert_alpha()
217.     logout_rect = logout.get_rect(topleft = (464,450))
218.
219.     username_disp = sub_font.render(username,True, "White")
220.     username_rect = username_disp.get_rect(topright = (948,8))
221.
222.     mambacreds = sub_font.render('Credits',True, "White")
223.     mambacreds_rect = mambacreds.get_rect(topright = (915,650))
224.
225.
226.
227.     while True:
228.
229.         screen.blit(background, (0,0))
230.         screen.blit(home,home_rect)
231.         screen.blit(username_disp,username_rect)
232.         screen.blit(games,games_rect)
233.         screen.blit(shop,shop_rect)
234.         screen.blit(logout,logout_rect)
235.         screen.blit(mambacreds,mambacreds_rect)
236.
237.         for event in pygame.event.get():
238.             if event.type == pygame.QUIT:
239.                 pygame.quit()
240.                 sys.exit()
241.
242.             if event.type ==pygame.MOUSEBUTTONDOWN:
243.                 if shop_rect.collidepoint(event.pos):
244.                     shop_main()
245.
246.             if event.type ==pygame.MOUSEBUTTONDOWN:
247.                 if games_rect.collidepoint(event.pos):
248.                     games_main()
249.
250.             if event.type ==pygame.MOUSEBUTTONDOWN:
251.                 if mambacreds_rect.collidepoint(event.pos):
252.                     mambacredits()
253.
254.             if event.type ==pygame.MOUSEBUTTONDOWN:
255.                 if logout_rect.collidepoint(event.pos):
256.                     username=''
257.                     password=''
258.                     startup()
259.
260.
261.

```

```

262.         pygame.display.update()
263.         clock.tick(60)
264.
265.     def mamba_login():
266.         global username, cartlist, cartno, merchandise, total, password
267.
268.         text_box1 = pygame.Rect(236,305,529,65)
269.         text_box2 = pygame.Rect(236,457,529,65)
270.
271.         activel = active2= False
272.         color1 = color2 = pygame.Color('black')
273.         clock = pygame.time.Clock()
274.         pygame.display.set_icon(ICON)
275.         pygame.display.set_caption("CHECKOUT")
276.         background = pygame.image.load('graphics\\mamba_login.png').convert_alpha()
277.
278.         home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
279.         home_rect = home.get_rect(topleft = (17,8))
280.
281.
282.         homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
283.         homehov_rect = homehov.get_rect(topleft = (11,8))
284.
285.         signup = pygame.image.load('graphics\\signup_button.png').convert_alpha()
286.         signup_rect = signup.get_rect(topleft = (387,349))
287.
288.         signin = pygame.image.load('graphics\\signin_button.png').convert_alpha()
289.         signin_rect = signin.get_rect(topleft = (387,246))
290.
291.         contquest = pygame.image.load('graphics\\contquest.png').convert_alpha()
292.         contquest_rect = contquest.get_rect(topleft = (406,445))
293.
294.
295.
296.
297.
298.     while True:
299.
300.         screen.blit(background, (0,0))
301.         screen.blit(home,home_rect)
302.         screen.blit(homehov,homehov_rect)
303.         screen.blit(contquest,contquest_rect)
304.         screen.blit(signup,signup_rect)
305.         screen.blit(signin,signin_rect)
306.
307.         for event in pygame.event.get():
308.             if event.type == pygame.QUIT:
309.                 pygame.quit()
310.                 sys.exit()
311.             if event.type ==pygame.MOUSEBUTTONDOWN:
312.                 if signup_rect.collidepoint(event.pos):
313.                     mamba_signup()
314.             if event.type ==pygame.MOUSEBUTTONDOWN:
315.                 if signin_rect.collidepoint(event.pos):
316.                     mamba_signin()
317.             if event.type ==pygame.MOUSEBUTTONDOWN:
318.                 if contquest_rect.collidepoint(event.pos):
319.                     username = "guest"
320.                     mamba_main()
321.             if event.type ==pygame.MOUSEBUTTONDOWN:
322.                 if home_rect.collidepoint(event.pos):
323.                     startup()
324.
325.
326.         pygame.display.update()
327.         clock.tick(60)
328.
329.

```

```

330. def mamba_signin():
331.     global username, cartlist, cartno, merchandise, total, password, credentials
332.
333.     username_input = ''
334.     password_input = ''
335.
336.     text_box1 = pygame.Rect(236, 305, 529, 65)
337.     text_box2 = pygame.Rect(236, 457, 529, 65)
338.
339.     active1 = active2 = False
340.     color1 = color2 = pygame.Color('black')
341.     clock = pygame.time.Clock()
342.     pygame.display.set_icon(ICON)
343.     pygame.display.set_caption("CHECKOUT")
344.     background = pygame.image.load('graphics\\signin.png').convert_alpha()
345.
346.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
347.     home_rect = home.get_rect(topleft = (17, 8))
348.
349.
350.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
351.     homehov_rect = homehov.get_rect(topleft = (11, 8))
352.
353.     confirm = pygame.image.load('graphics\\confirm.png').convert_alpha()
354.     confirm_rect = confirm.get_rect(topleft = (365, 549))
355.
356.     unsuccessful = guest_font.render("Username or Password
Incorrect", True, "Red")
357.     unsuccessful_rect = unsuccessful.get_rect(topleft = (280, 643))
358.
359.
360.
361.     while True:
362.
363.         screen.blit(background, (0, 0))
364.         screen.blit(home, home_rect)
365.         screen.blit(confirm, confirm_rect)
366.
367.         for event in pygame.event.get():
368.             if event.type == pygame.QUIT:
369.                 pygame.quit()
370.                 sys.exit()
371.             if event.type == pygame.MOUSEBUTTONDOWN:
372.                 if home_rect.collidepoint(event.pos):
373.                     mamba_login
374.
375.             if event.type == pygame.MOUSEBUTTONDOWN:
376.                 if homehov_rect.collidepoint(event.pos):
377.                     mamba_login()
378.
379.             if event.type == pygame.MOUSEBUTTONDOWN:
380.                 if confirm_rect.collidepoint(event.pos):
381.                     username = username_input.lower()
382.                     password = password_input
383.                     if username != '' and password != '':
384.                         cursor.execute("SELECT * FROM CREDENTIALS")
385.                         usernames = []
386.                         a = cursor.fetchall()
387.                         for i in a:
388.                             if i[0] == username:
389.                                 if i[1] == password:
390.                                     credentials.append(i)
391.                                     mamba_main()
392.                             else:
393.                                 screen.blit(unsuccessful, unsuccessful_rect)
394.                                 usernames.append(i[0])
395.                                 if username not in usernames:
396.                                     screen.blit(unsuccessful, unsuccessful_rect)

```

```

397.         else:
398.             screen.blit(unsuccesful,unsuccesful_rect)
399.
400.
401.     if event.type == pygame.KEYDOWN:
402.         if event.key == pygame.K_RETURN:
403.             username = username_input.lower()
404.             password = password_input
405.             if username != '' and password != '':
406.                 cursor.execute("SELECT * FROM CREDENTIALS")
407.                 usernames = []
408.                 a = cursor.fetchall()
409.                 for i in a:
410.                     if i[0] == username:
411.                         if i[1] ==password:
412.                             mamba_main()
413.                         else:
414.                             screen.blit(unsuccesful,unsuccesful_rect)
415.                             usernames.append(i[0])
416.                             if username not in usernames:
417.                                 screen.blit(unsuccesful,unsuccesful_rect)
418.                             else:
419.                                 screen.blit(unsuccesful,unsuccesful_rect)
420.
421.
422.     if event.type == pygame.MOUSEBUTTONDOWN:
423.         if text_box1.collidepoint(event.pos):
424.             active1 = True
425.         else:
426.             active1 = False
427.     if event.type == pygame.KEYDOWN:
428.         if active1:
429.             if event.key == pygame.K_BACKSPACE:
430.                 username_input = username_input[:-1]
431.             else:
432.                 username_input += event.unicode
433.
434.     if event.type == pygame.MOUSEBUTTONDOWN:
435.         if text_box2.collidepoint(event.pos):
436.             active2 = True
437.         else:
438.             active2 = False
439.     if event.type == pygame.KEYDOWN:
440.         if active2:
441.             if event.key == pygame.K_BACKSPACE:
442.                 password_input = password_input[:-1]
443.             else:
444.                 password_input += event.unicode
445.
446.
447.     if active1 :
448.         color1 = pygame.Color('white')
449.     else:
450.         color1 = pygame.Color('black')
451.
452.     if active2:
453.         color2 = pygame.Color('white')
454.     else:
455.         color2 = pygame.Color('black')
456.
457.
458.     pygame.draw.rect(screen,color1, text_box1,4)
459.     pygame.draw.rect(screen,color2, text_box2,4)
460.
461.     passworddisp = ('*' *len(password_input))
462.
463.     username = text_input_font.render(username_input,True,'white')
464.     password = text_input_font.render(passworddisp,True,'white')

```

```

465.
466.
467.
468.
469.         screen.blit(username, (text_box1.x , text_box1.y))
470.         screen.blit(password, (text_box2.x , text_box2.y))
471.
472.         pygame.display.update()
473.         clock.tick(10)
474.
475. def mamba_signup():
476.     global username, cartlist, cartno, merchandise, total, password
477.
478.     username_input = ''
479.     password_input = ''
480.
481.     text_box1 = pygame.Rect(236,305,529,65)
482.     text_box2 = pygame.Rect(236,457,529,65)
483.
484.     active1 = active2= False
485.     color1 = color2 = pygame.Color('black')
486.     clock = pygame.time.Clock()
487.     pygame.display.set_icon(ICON)
488.     pygame.display.set_caption("CHECKOUT")
489.     background = pygame.image.load('graphics\\signup.png').convert_alpha()
490.
491.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
492.     home_rect = home.get_rect(topleft = (17,8))
493.
494.
495.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
496.     homehov_rect = homehov.get_rect(topleft = (11,8))
497.
498.     confirm = pygame.image.load('graphics\\confirm.png').convert_alpha()
499.     confirm_rect = confirm.get_rect(topleft = (365,549))
500.
501.     unsuccessful = guest_font.render("Username or Password
Incorrect", True, "Red")
502.     unsuccessful_rect = unsuccessful.get_rect(topleft = (280,643))
503.
504.     usernametaken = guest_font.render("username already taken", True, "Red")
505.     usernametaken_rect = usernametaken.get_rect(topleft = (360,643))
506.
507.
508.
509.     while True:
510.
511.         screen.blit(background, (0,0))
512.         screen.blit(home,home_rect)
513.         screen.blit(confirm,confirm_rect)
514.
515.         for event in pygame.event.get():
516.             if event.type == pygame.QUIT:
517.                 pygame.quit()
518.                 sys.exit()
519.             if event.type ==pygame.MOUSEBUTTONDOWN:
520.                 if home_rect.collidepoint(event.pos):
521.                     mamba_login()
522.             if event.type ==pygame.MOUSEBUTTONDOWN:
523.                 if homehov_rect.collidepoint(event.pos):
524.                     mamba_login()
525.
526.             if event.type ==pygame.MOUSEBUTTONDOWN:
527.                 if confirm_rect.collidepoint(event.pos):
528.                     username = username_input.lower()
529.                     password = password_input
530.                     if username != '' and password != '':
531.                         cursor.execute("SELECT * FROM CREDENTIALS")

```

```

532.             usernames = []
533.             a = cursor.fetchall()
534.             mysql.commit()
535.             if a == []:
536.                 cursor.execute("INSERT INTO
CREDENTIALS (USERNAME,PASSWORD) VALUES(%s,%s)", (username,password))
537.                 mysql.commit()
538.                 cursor.execute("INSERT INTO SCORES (USERNAME)
VALUES(%s)", (username,))
539.                 mysql.commit()
540.                 mamba_main()
541.                 for i in a:
542.                     if i[0] == username or username == 'guest':
543.                         screen.blit( usernametaken, usernametaken_rect)
544.                     else:
545.                         cursor.execute("INSERT INTO
CREDENTIALS (USERNAME,PASSWORD) VALUES(%s,%s)", (username,password))
546.                         mysql.commit()
547.                         cursor.execute("INSERT INTO SCORES (USERNAME)
VALUES(%s)", (username,))
548.                         mysql.commit()
549.                         mamba_main()
550.                 else:
551.                     screen.blit( unsuccessful, unsuccessful_rect)
552.
553.
554.
555.             if event.type == pygame.MOUSEBUTTONDOWN:
556.                 if text_box1.collidepoint(event.pos):
557.                     active1 = True
558.                 else:
559.                     active1 = False
560.             if event.type == pygame.KEYDOWN:
561.                 if active1:
562.                     if event.key == pygame.K_BACKSPACE:
563.                         username_input = username_input[:-1]
564.                     else:
565.                         username_input += event.unicode
566.
567.                 if event.type == pygame.MOUSEBUTTONDOWN:
568.                     if text_box2.collidepoint(event.pos):
569.                         active2 = True
570.                     else:
571.                         active2 = False
572.             if event.type == pygame.KEYDOWN:
573.                 if active2:
574.                     if event.key == pygame.K_BACKSPACE:
575.                         password_input = password_input[:-1]
576.                     else:
577.                         password_input += event.unicode
578.
579.
580.                 if active1 :
581.                     color1 = pygame.Color('white')
582.                 else:
583.                     color1 = pygame.Color('black')
584.
585.                 if active2:
586.                     color2 = pygame.Color('white')
587.                 else:
588.                     color2 = pygame.Color('black')
589.
590.                 passworddisp = ('*' * len(password_input))
591.                 pygame.draw.rect(screen,color1, text_box1,4)
592.                 pygame.draw.rect(screen,color2, text_box2,4)
593.
594.
595.                 username = text_input_font.render(username_input,True,'white')

```

```

596.         password = text_input_font.render(passworddisp,True,'white')
597.
598.
599.         screen.blit(username, (text_box1.x , text_box1.y))
600.         screen.blit(password, (text_box2.x , text_box2.y))
601.
602.         pygame.display.update()
603.         clock.tick(10)
604.
605.     def success():
606.         global username, cartlist, cartno, merchandise
607.         cartno = 0
608.         cartlist = {101:0,102:0,201:0,202:0,203:0,301:0,302:0,303:0}
609.         pygame.display.set_icon(ICON)
610.         pygame.display.set_caption("CHECKOUT")
611.         background = pygame.image.load('graphics\\end_screen.png').convert_alpha()
612.
613.         home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
614.         home_rect = home.get_rect(topleft = (17,8))
615.
616.         cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
617.         cartmain_rect = cartmain.get_rect(topleft = (882,8))
618.
619.         homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
620.         homehov_rect = homehov.get_rect(topleft = (11,8))
621.
622.         home_big = pygame.image.load('graphics\\home.png').convert_alpha()
623.         home_big_rect = home_big.get_rect(topleft = (368,565))
624.
625.
626.
627.         username_disp = sub_font.render(username,True, "White")
628.         username_rect = username_disp.get_rect(topright = (872,7))
629.
630.         cartnumber = number_font.render(str(cartno),True,"White")
631.         cartnumber_rect = cartnumber.get_rect(center = (902,26))
632.
633.         while True:
634.
635.             screen.blit(background,(0,0))
636.             screen.blit(home,home_rect)
637.             screen.blit(cartmain,cartmain_rect)
638.             screen.blit(username_disp,username_rect)
639.             screen.blit(cartnumber,cartnumber_rect)
640.             screen.blit(home_big,home_big_rect)
641.
642.             for event in pygame.event.get():
643.                 if event.type == pygame.QUIT:
644.                     pygame.quit()
645.                     sys.exit()
646.                 if event.type ==pygame.MOUSEBUTTONDOWN:
647.                     if home_rect.collidepoint(event.pos):
648.                         shop_main()
649.                 if event.type ==pygame.MOUSEBUTTONDOWN:
650.                     if home_big_rect.collidepoint(event.pos):
651.                         shop_main()
652.
653.
654.             pygame.display.update()
655.             clock.tick(60)
656.
657.     def checkout_address():
658.         global username, cartlist, cartno, merchandise, total
659.
660.         email_input = ''
661.         address_input = ''
662.         phoneno_input = ''
663.

```

```

664.     text_box1 = pygame.Rect(87,302,529,65)
665.     text_box2 = pygame.Rect(87,423,529,65)
666.     text_box3 = pygame.Rect(87,547,529,65)
667.
668.     active1 = active2= active3= False
669.     color1 = color2 = color3 = pygame.Color('black')
670.     clock = pygame.time.Clock()
671.     pygame.display.set_icon(ICON)
672.     pygame.display.set_caption("CHECKOUT")
673.     background =
pygame.image.load('graphics\\checkout_address.png').convert_alpha()
674.
675.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
676.     home_rect = home.get_rect(topleft = (17,8))
677.
678.     cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
679.     cartmain_rect = cartmain.get_rect(topleft = (882,8))
680.
681.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
682.     homehov_rect = homehov.get_rect(topleft = (11,8))
683.
684.     confirm = pygame.image.load('graphics\\confirm.png').convert_alpha()
685.     confirm_rect = confirm.get_rect(topleft = (686,534))
686.
687.     savedet = pygame.image.load('graphics\\savedetails.png').convert_alpha()
688.     savedet_rect = savedet.get_rect(topleft = (686,412))
689.
690.
691.     username_disp = sub_font.render(username,True, "White")
692.     username_rect = username_disp.get_rect(topright = (872,7))
693.
694.     cartnumber = number_font.render(str(cartno),True,"White")
695.     cartnumber_rect = cartnumber.get_rect(center = (902,26))
696.
697.     price = finalprice_font.render(str(total),True,"White")
698.     price_rect = price.get_rect(topleft = (689,292))
699.
700.
701.
702.
703.     while True:
704.
705.         screen.blit(background,(0,0))
706.         screen.blit(home,home_rect)
707.         screen.blit(cartmain,cartmain_rect)
708.         screen.blit(username_disp,username_rect)
709.         screen.blit(cartnumber,cartnumber_rect)
710.         screen.blit(price,price_rect)
711.         screen.blit(confirm,confirm_rect)
712.         if username != 'guest':
713.             screen.blit(savedet,savedet_rect)
714.
715.         for event in pygame.event.get():
716.             if event.type == pygame.QUIT:
717.                 pygame.quit()
718.                 sys.exit()
719.             if event.type ==pygame.MOUSEBUTTONDOWN:
720.                 if home_rect.collidepoint(event.pos):
721.                     shop_main()
722.             if event.type ==pygame.MOUSEBUTTONDOWN:
723.                 if confirm_rect.collidepoint(event.pos):
724.                     success()
725.             if event.type ==pygame.MOUSEBUTTONDOWN:
726.                 if savedet_rect.collidepoint(event.pos):
727.                     if username != 'guest':
728.                         cursor.execute("UPDATE CREDENTIALS SET ADDRESS = %s,
EMAIL = %s, PHONENO = %s, CCNO = %s,EXPIRYDATE = %s, CVV = %s WHERE USERNAME = %s

```

```

", (address_input, email_input, int(phoneno_input), creditno_input, creditexpiry_input, creditcvv_input, username))
729.         mysql.commit()
730.
731.         if event.type == pygame.MOUSEBUTTONDOWN:
732.             if text_box1.collidepoint(event.pos):
733.                 active1 = True
734.             else:
735.                 active1 = False
736.         if event.type == pygame.KEYDOWN:
737.             if active1:
738.                 if event.key == pygame.K_BACKSPACE:
739.                     address_input = address_input[:-1]
740.                 else:
741.                     address_input += event.unicode
742.
743.             if event.type == pygame.MOUSEBUTTONDOWN:
744.                 if text_box2.collidepoint(event.pos):
745.                     active2 = True
746.                 else:
747.                     active2 = False
748.             if event.type == pygame.KEYDOWN:
749.                 if active2:
750.                     if event.key == pygame.K_BACKSPACE:
751.                         email_input = email_input[:-1]
752.                     else:
753.                         email_input += event.unicode
754.
755.             if event.type == pygame.MOUSEBUTTONDOWN:
756.                 if text_box3.collidepoint(event.pos):
757.                     active3 = True
758.                 else:
759.                     active3 = False
760.             if event.type == pygame.KEYDOWN:
761.                 if active3:
762.                     if event.key == pygame.K_BACKSPACE:
763.                         phoneno_input = phoneno_input[:-1]
764.                     else:
765.                         phoneno_input += event.unicode
766.
767.
768.         if active1 :
769.             color1 = pygame.Color('white')
770.         else:
771.             color1 = pygame.Color('black')
772.
773.         if active2:
774.             color2 = pygame.Color('white')
775.         else:
776.             color2 = pygame.Color('black')
777.
778.         if active3:
779.             color3 = pygame.Color('white')
780.         else:
781.             color3 = pygame.Color('black')
782.
783.         pygame.draw.rect(screen, color1, text_box1, 4)
784.         pygame.draw.rect(screen, color2, text_box2, 4)
785.         pygame.draw.rect(screen, color3, text_box3, 4)
786.
787.         address = text_input_font.render(address_input, True, 'white')
788.         email = text_input_font.render(email_input, True, 'white')
789.         phoneno = text_input_font.render(phoneno_input, True, 'white')
790.
791.         screen.blit(address, (text_box1.x , text_box1.y))
792.         screen.blit(email, (text_box2.x , text_box2.y))
793.         screen.blit(phoneno, (text_box3.x , text_box3.y))
794.

```

```

795.         pygame.display.update()
796.         clock.tick(60)
797.
798.
799.
800. def saved_checkout():
801.     global username, cartlist, cartno, merchandise, total
802.
803.
804.     clock = pygame.time.Clock()
805.     pygame.display.set_icon(ICON)
806.     pygame.display.set_caption("CHECKOUT")
807.     background =
pygame.image.load('graphics\\savecheckout.png').convert_alpha()
808.
809.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
810.     home_rect = home.get_rect(topleft = (17,8))
811.
812.     cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
813.     cartmain_rect = cartmain.get_rect(topleft = (882,8))
814.
815.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
816.     homehov_rect = homehov.get_rect(topleft = (11,8))
817.
818.     yes = pygame.image.load('graphics\\saveyes.png').convert_alpha()
819.     yes_rect = yes.get_rect(topleft = (191,565))
820.
821.     no = pygame.image.load('graphics\\saveno.png').convert_alpha()
822.     no_rect = no.get_rect(topleft = (540,567))
823.
824.
825.
826.     username_disp = sub_font.render(username,True, "White")
827.     username_rect = username_disp.get_rect(topright = (872,7))
828.
829.     cartnumber = number_font.render(str(cartno),True,"White")
830.     cartnumber_rect = cartnumber.get_rect(center = (902,26))
831.
832.     price = finalprice_font.render(str(total),True,"White")
833.     price_rect = price.get_rect(topleft = (689,292))
834.
835.
836.
837.
838. while True:
839.
840.     screen.blit(background,(0,0))
841.     screen.blit(home,home_rect)
842.     screen.blit(cartmain,cartmain_rect)
843.     screen.blit(username_disp,username_rect)
844.     screen.blit(cartnumber,cartnumber_rect)
845.     screen.blit(price,price_rect)
846.     screen.blit(yes,yes_rect)
847.     screen.blit(no,no_rect)
848.
849.     for event in pygame.event.get():
850.         if event.type == pygame.QUIT:
851.             pygame.quit()
852.             sys.exit()
853.         if event.type ==pygame.MOUSEBUTTONDOWN:
854.             if home_rect.collidepoint(event.pos):
855.                 shop_main()
856.         if event.type ==pygame.MOUSEBUTTONDOWN:
857.             if yes_rect.collidepoint(event.pos):
858.                 success()
859.         if event.type ==pygame.MOUSEBUTTONDOWN:
860.             if no_rect.collidepoint(event.pos):
861.                 checkout_credit()

```

```

862.
863.
864.     pygame.display.update()
865.     clock.tick(60)
866.
867.
868. def checkout_credit():
869.     global username, cartlist, cartno, merchandise, total, creditno_input,
creditexpiry_input, creditcvv_input
870.
871.     creditno_input = ''
872.     creditexpiry_input = ''
873.     creditcvv_input = ''
874.
875.     text_box1 = pygame.Rect(87,302,529,65)
876.     text_box2 = pygame.Rect(87,423,529,65)
877.     text_box3 = pygame.Rect(87,547,529,65)
878.
879.     active1 = active2= active3= False
880.     color1 = color2 = color3 = pygame.Color('black')
881.     clock = pygame.time.Clock()
882.     pygame.display.set_icon(ICON)
883.     pygame.display.set_caption("CHECKOUT")
884.     background =
pygame.image.load('graphics\\checkout_credit.png').convert_alpha()
885.
886.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
887.     home_rect = home.get_rect(topleft = (17,8))
888.
889.     cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
890.     cartmain_rect = cartmain.get_rect(topleft = (882,8))
891.
892.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
893.     homehov_rect = home.get_rect(topleft = (11,8))
894.
895.     placeorder = pygame.image.load('graphics\\place_order.png').convert_alpha()
896.     placeorder_rect = placeorder.get_rect(topleft = (686,477))
897.
898.
899.     username_disp = sub_font.render(username,True, "White")
900.     username_rect = username_disp.get_rect(topright = (872,7))
901.
902.     cartnumber = number_font.render(str(cartno),True,"White")
903.     cartnumber_rect = cartnumber.get_rect(center = (902,26))
904.
905.     price = finalprice_font.render(str(total),True,"White")
906.     price_rect = price.get_rect(topleft = (689,292))
907.
908.     creditno = text_input_font.render(str(creditno_input),True,"White")
909.     creditno_rect = creditno.get_rect(topleft = (119,302))
910.
911.
912.     while True:
913.
914.         screen.blit(background,(0,0))
915.         screen.blit(home,home_rect)
916.         screen.blit(cartmain,cardmain_rect)
917.         screen.blit(username_disp,username_rect)
918.         screen.blit(cartnumber,cartnumber_rect)
919.         screen.blit(price,price_rect)
920.         screen.blit(placeorder,placeorder_rect)
921.
922.         for event in pygame.event.get():
923.             if event.type == pygame.QUIT:
924.                 pygame.quit()
925.                 sys.exit()
926.             if event.type ==pygame.MOUSEBUTTONDOWN:
927.                 if home_rect.collidepoint(event.pos):

```

```

928.         shop_main()
929.     if event.type == pygame.MOUSEBUTTONDOWN:
930.         if placeorder_rect.collidepoint(event.pos):
931.             checkout_address()
932.
933.     if event.type == pygame.MOUSEBUTTONDOWN:
934.         if text_box1.collidepoint(event.pos):
935.             active1 = True
936.         else:
937.             active1 = False
938.     if event.type == pygame.KEYDOWN:
939.         if active1:
940.             if event.key == pygame.K_BACKSPACE:
941.                 creditno_input = creditno_input[:-1]
942.             else:
943.                 creditno_input += event.unicode
944.
945.     if event.type == pygame.MOUSEBUTTONDOWN:
946.         if text_box2.collidepoint(event.pos):
947.             active2 = True
948.         else:
949.             active2 = False
950.     if event.type == pygame.KEYDOWN:
951.         if active2:
952.             if event.key == pygame.K_BACKSPACE:
953.                 creditexpiry_input = creditexpiry_input[:-1]
954.             else:
955.                 creditexpiry_input += event.unicode
956.
957.     if event.type == pygame.MOUSEBUTTONDOWN:
958.         if text_box3.collidepoint(event.pos):
959.             active3 = True
960.         else:
961.             active3 = False
962.     if event.type == pygame.KEYDOWN:
963.         if active3:
964.             if event.key == pygame.K_BACKSPACE:
965.                 creditcvv_input = creditcvv_input[:-1]
966.             else:
967.                 creditcvv_input += event.unicode
968.
969.
970.     if active1 :
971.         color1 = pygame.Color('white')
972.     else:
973.         color1 = pygame.Color('black')
974.
975.     if active2:
976.         color2 = pygame.Color('white')
977.     else:
978.         color2 = pygame.Color('black')
979.
980.     if active3:
981.         color3 = pygame.Color('white')
982.     else:
983.         color3 = pygame.Color('black')
984.
985.     pygame.draw.rect(screen,color1, text_box1,4)
986.     pygame.draw.rect(screen,color2, text_box2,4)
987.     pygame.draw.rect(screen,color3, text_box3,4)
988.
989.     creditno = text_input_font.render(creditno_input,True,'white')
990.     creditexpiry = text_input_font.render(creditexpiry_input,True,'white')
991.     creditcvv = text_input_font.render(creditcvv_input,True,'white')
992.
993.     screen.blit(creditno, (text_box1.x , text_box1.y))
994.     screen.blit(creditexpiry, (text_box2.x , text_box2.y))
995.     screen.blit(creditcvv, (text_box3.x , text_box3.y))

```

```

996.
997.         pygame.display.update()
998.         clock.tick(60)
999.
1000. def emptybill():
1001.     global username, cartlist, cartno, merchandise
1002.     finalbill=[]
1003.     pygame.display.set_icon(ICON)
1004.     pygame.display.set_caption("CHECKOUT")
1005.     background =
pygame.image.load('graphics\\checkoutbill.png').convert_alpha()
1006.
1007.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
1008.     home_rect = home.get_rect(topleft = (17,8))
1009.
1010.     cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
1011.     cartmain_rect = cartmain.get_rect(topleft = (882,8))
1012.
1013.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
1014.     homehov_rect = home.get_rect(topleft = (11,8))
1015.
1016.     checkout = pygame.image.load('graphics\\checkout.png').convert_alpha()
1017.     checkout_rect = home.get_rect(topleft = (626,635))
1018.
1019.     clearcart = pygame.image.load('graphics\\clearcart.png').convert_alpha()
1020.     clearcart_rect = home.get_rect(topleft = (774,203))
1021.
1022.
1023.     username_disp = sub_font.render(username,True, "White")
1024.     username_rect = username_disp.get_rect(topright = (872,7))
1025.
1026.     cartnumber = number_font.render(str(cartno),True,"White")
1027.     cartnumber_rect = cartnumber.get_rect(center = (902,26))
1028.
1029.     cartitem_top = 285
1030.     total = 0
1031.     a = x = 0
1032.
1033.
1034.
1035.     while True:
1036.
1037.         screen.blit(background,(0,0))
1038.         screen.blit(home,home_rect)
1039.         screen.blit(cartmain,cardmain_rect)
1040.         screen.blit(username_disp,username_rect)
1041.         screen.blit(cartnumber,cardnumber_rect)
1042.         screen.blit(checkout,checkout_rect)
1043.         screen.blit(clearcart,clearcart_rect)
1044.
1045.         for event in pygame.event.get():
1046.             if event.type == pygame.QUIT:
1047.                 pygame.quit()
1048.                 sys.exit()
1049.             if event.type ==pygame.MOUSEBUTTONDOWN:
1050.                 if home_rect.collidepoint(event.pos):
1051.                     shop_main()
1052.             if event.type ==pygame.MOUSEMOTION:
1053.                 if home_rect.collidepoint(event.pos):
1054.                     screen.blit(homehov,homehov_rect)
1055.             if event.type ==pygame.MOUSEBUTTONDOWN:
1056.                 if clearcart_rect.collidepoint(event.pos):
1057.                     cartno = 0
1058.                     for i in cartlist:
1059.                         cartlist[i] = 0
1060.
1061.
1062.

```

```

1063.         pygame.display.update()
1064.         clock.tick(60)
1065.
1066.
1067. def checkout_bill():
1068.     global username, cartlist, cartno, merchandise, total, credentials
1069.     finalbill=[]
1070.     pygame.display.set_icon(ICON)
1071.     pygame.display.set_caption("CHECKOUT")
1072.     background =
pygame.image.load('graphics\\checkoutbill.png').convert_alpha()
1073.
1074.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
1075.     home_rect = home.get_rect(topleft = (17,8))
1076.
1077.     cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
1078.     cartmain_rect = cartmain.get_rect(topleft = (882,8))
1079.
1080.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
1081.     homehov_rect = home.get_rect(topleft = (11,8))
1082.
1083.     checkout = pygame.image.load('graphics\\checkout.png').convert_alpha()
1084.     checkout_rect = home.get_rect(topleft = (626,635))
1085.
1086.     clearcart = pygame.image.load('graphics\\clearcart.png').convert_alpha()
1087.     clearcart_rect = home.get_rect(topleft = (774,203))
1088.
1089.     username_disp = sub_font.render(username,True, "White")
1090.     username_rect = username_disp.get_rect(topright = (872,7))
1091.
1092.     cartnumber = number_font.render(str(cartno),True,"White")
1093.     cartnumber_rect = cartnumber.get_rect(center = (902,26))
1094.
1095.     cartitem_top = 285
1096.     total = 0
1097.     w8 = x = 0
1098.
1099.
1100.
1101. while True:
1102.
1103.     screen.blit(background,(0,0))
1104.     screen.blit(home,home_rect)
1105.     screen.blit(cartmain,cardmain_rect)
1106.     screen.blit(username_disp,username_rect)
1107.     screen.blit(cartnumber,cardnumber_rect)
1108.     screen.blit(checkout,checkout_rect)
1109.     screen.blit(clearcart,clearcart_rect)
1110.
1111.     for event in pygame.event.get():
1112.         if event.type == pygame.QUIT:
1113.             pygame.quit()
1114.             sys.exit()
1115.         if event.type ==pygame.MOUSEBUTTONDOWN:
1116.             if home_rect.collidepoint(event.pos):
1117.                 shop_main()
1118.         if event.type ==pygame.MOUSEMOTION:
1119.             if home_rect.collidepoint(event.pos):
1120.                 screen.blit(homehov,homehov_rect)
1121.         if event.type ==pygame.MOUSEBUTTONDOWN:
1122.             if clearcart_rect.collidepoint(event.pos):
1123.                 cartno = 0
1124.                 for i in cartlist:
1125.                     cartlist[i] = 0
1126.                 emptybill()
1127.         if event.type ==pygame.MOUSEBUTTONDOWN:
1128.             if checkout_rect.collidepoint(event.pos):
1129.                 if username != 'guest':

```

```

1130.             cursor.execute("SELECT * FROM CREDENTIALS WHERE
USERNAME = %s", (username,))
1131.             a = cursor.fetchall()
1132.             if a[0][2] != None:
1133.                 saved_checkout()
1134.             else:
1135.                 checkout_credit()
1136.             else:
1137.                 checkout_credit()
1138.
1139.
1140.         #printing the items in the bill
1141.         while w8 != 8:
1142.             for i in cartlist:
1143.                 if cartlist[i] != 0 :
1144.
1145.                     cartitemno = i
1146.                     cartitemname = merchandise[i][0]
1147.                     cartitemprice = merchandise[i][1]
1148.                     cartitemqty = cartlist[i]
1149.                     cartitemtotal = merchandise[i][1]*cartlist[i]
1150.
1151. finalbill.append([cartitemno, cartitemname, cartitemprice, cartitemqty, cartitemtotal])
1152.             total += cartitemtotal
1153.             w8 +=1
1154.
1155.
1156.         if x != len(finalbill):
1157.             for i in finalbill:
1158.                 cartitemno = bill_font.render(str(i[0]), True, "White")
1159.                 cartitemno_rect = cartitemno.get_rect(center =
(75, cartitem_top))
1160.
1161.                 cartitemname = bill_font.render(str(i[1]), True, "White")
1162.                 cartitemname_rect = cartitemname.get_rect(topleft =
(193, cartitem_top - 20))
1163.
1164.                 cartitemprice = bill_font.render(str(i[2]), True, "White")
1165.                 cartitemprice_rect = cartitemprice.get_rect(center =
(516, cartitem_top))
1166.
1167.                 cartitemqty = bill_font.render(str(i[3]), True, "White")
1168.                 cartitemqty_rect = cartitemqty.get_rect(center =
(631, cartitem_top))
1169.
1170.                 cartitemtotal = bill_font.render(str(i[4]), True, "White")
1171.                 cartitemtotal_rect = cartitemtotal.get_rect(center =
(881, cartitem_top))
1172.
1173.                 totalprice = total_font.render(str(total), True, "White")
1174.                 totalprice_rect = totalprice.get_rect(topleft = (456, 627))
1175.
1176.                 dirhams = total_font.render("AED", True, "White")
1177.                 dirhams_rect = totalprice.get_rect(topleft = (538, 627))
1178.
1179.
1180.
1181.
1182.
1183.                 screen.blit(cartitemno, cartitemno_rect)
1184.                 screen.blit(cartitemname, cartitemname_rect)
1185.                 screen.blit(cartitemprice, cartitemprice_rect)
1186.                 screen.blit(cartitemqty, cartitemqty_rect)
1187.                 screen.blit(cartitemtotal, cartitemtotal_rect)
1188.                 screen.blit(totalprice, totalprice_rect)
1189.                 screen.blit(dirhams, dirhams_rect)
1190.             pygame.display.update()

```

```

1191.
1192.
1193.
1194.
1195.         x+=1
1196.         cartitem_top +=45
1197.
1198.
1199.         clock.tick(60)
1200.
1201.
1202.
1203.
1204.
1205.
1206. def shop_shirt():
1207.     global username, cartlist, cartno, merchandise
1208.     pygame.display.set_icon(ICON)
1209.     pygame.display.set_caption("SHIRT SHOP")
1210.     background = pygame.image.load('graphics\\shopshirts.png').convert_alpha()
1211.
1212.     cart1 = pygame.image.load('graphics\\addtocart.png').convert_alpha()
1213.     cart1_rect = cart1.get_rect(topleft = (278,622))
1214.
1215.     cart2 = pygame.image.load('graphics\\addtocart.png').convert_alpha()
1216.     cart2_rect = cart2.get_rect(topleft = (566,622))
1217.
1218.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
1219.     home_rect = home.get_rect(topleft = (17,8))
1220.
1221.     cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
1222.     cartmain_rect = cartmain.get_rect(topleft = (882,8))
1223.
1224.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
1225.     homehov_rect = home.get_rect(topleft = (11,8))
1226.
1227.     username_disp = sub_font.render(username,True, "White")
1228.     username_rect = username_disp.get_rect(topright = (872,7))
1229.
1230.     cartnumber = number_font.render(str(cartno),True,"White")
1231.     cartnumber_rect = cartnumber.get_rect(center = (902,26))
1232.
1233.
1234.
1235.
1236. while True:
1237.
1238.     screen.blit(background,(0,0))
1239.     screen.blit(home,home_rect)
1240.     screen.blit(cart1,cart1_rect)
1241.     screen.blit(cart2,cart2_rect)
1242.     screen.blit(cartmain,cartmain_rect)
1243.     screen.blit(username_disp,username_rect)
1244.     screen.blit(cartnumber,cartnumber_rect)
1245.
1246.     for event in pygame.event.get():
1247.         if event.type == pygame.QUIT:
1248.             pygame.quit()
1249.             sys.exit()
1250.         if event.type ==pygame.MOUSEBUTTONDOWN:
1251.             if home_rect.collidepoint(event.pos):
1252.                 shop_main()
1253.         if event.type ==pygame.MOUSEMOTION:
1254.             if home_rect.collidepoint(event.pos):
1255.                 screen.blit(homehov,homehov_rect)
1256.         if event.type ==pygame.MOUSEBUTTONDOWN:
1257.             if cart1_rect.collidepoint(event.pos):
1258.                 cartno+=1

```

```

1259.         cartlist[101]+=1
1260.         if event.type ==pygame.MOUSEBUTTONDOWN:
1261.             if cart2_rect.collidepoint(event.pos):
1262.                 cartno+=1
1263.                 cartlist[102]+=1
1264.         if event.type ==pygame.MOUSEBUTTONDOWN:
1265.             if cartmain_rect.collidepoint(event.pos):
1266.                 checkout_bill()
1267.
1268.
1269.
1270.
1271.
1272.         pygame.display.update()
1273.         clock.tick(60)
1274.
1275. def shop_energy():
1276.     global username, cartlist, cartno, merchandise
1277.     pygame.display.set_icon(ICON)
1278.     pygame.display.set_caption("ENERGY SHOP")
1279.     background = pygame.image.load('graphics\\shopenergy.png').convert_alpha()
1280.
1281.     cart1 = pygame.image.load('graphics\\addtocart.png').convert_alpha()
1282.     cart1_rect = cart1.get_rect(topleft = (135,613))
1283.
1284.     cart2 = pygame.image.load('graphics\\addtocart.png').convert_alpha()
1285.     cart2_rect = cart2.get_rect(topleft = (423,613))
1286.
1287.     cart3 = pygame.image.load('graphics\\addtocart.png').convert_alpha()
1288.     cart3_rect = cart3.get_rect(topleft = (720,613))
1289.
1290.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
1291.     home_rect = home.get_rect(topleft = (17,8))
1292.
1293.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
1294.     homehov_rect = home.get_rect(topleft = (11,8))
1295.
1296.     cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
1297.     cartmain_rect = cartmain.get_rect(topleft = (882,8))
1298.
1299.     username_disp = sub_font.render(username,True, "White")
1300.     username_rect = username_disp.get_rect(topright = (872,7))
1301.
1302.     cartnumber = number_font.render(str(cartno),True,"White")
1303.     cartnumber_rect = cartnumber.get_rect(center = (902,26))
1304.
1305.     while True:
1306.
1307.         screen.blit(background,(0,0))
1308.         screen.blit(home,home_rect)
1309.         screen.blit(cart1,card1_rect)
1310.         screen.blit(cart2,card2_rect)
1311.         screen.blit(cart3,card3_rect)
1312.         screen.blit(cartmain,cardmain_rect)
1313.         screen.blit(username_disp,username_rect)
1314.         screen.blit(cartnumber,cardnumber_rect)
1315.
1316.         for event in pygame.event.get():
1317.             if event.type == pygame.QUIT:
1318.                 pygame.quit()
1319.                 sys.exit()
1320.             if event.type ==pygame.MOUSEBUTTONDOWN:
1321.                 if home_rect.collidepoint(event.pos):
1322.                     shop_main()
1323.             if event.type ==pygame.MOUSEMOTION:
1324.                 if home_rect.collidepoint(event.pos):
1325.                     screen.blit(homehov,homehov_rect)
1326.             if event.type ==pygame.MOUSEBUTTONDOWN:

```

```

1327.         if cart1_rect.collidepoint(event.pos):
1328.             cartno+=1
1329.             cartlist[301]+=1
1330.         if event.type ==pygame.MOUSEBUTTONDOWN:
1331.             if cart2_rect.collidepoint(event.pos):
1332.                 cartno+=1
1333.                 cartlist[302]+=1
1334.         if event.type ==pygame.MOUSEBUTTONDOWN:
1335.             if cart3_rect.collidepoint(event.pos):
1336.                 cartno+=1
1337.                 cartlist[303]+=1
1338.         if event.type ==pygame.MOUSEBUTTONDOWN:
1339.             if cartmain_rect.collidepoint(event.pos):
1340.                 checkout_bill()
1341.
1342.
1343.
1344.
1345.         pygame.display.update()
1346.         clock.tick(60)
1347.
1348.     def shop_mugs():
1349.         global username, cartlist, cartno, merchandise
1350.         pygame.display.set_icon(ICON)
1351.         pygame.display.set_caption("MUG SHOP")
1352.         background = pygame.image.load('graphics\\shopmugs.png').convert_alpha()
1353.
1354.         cart1 = pygame.image.load('graphics\\addtocart.png').convert_alpha()
1355.         cart1_rect = cart1.get_rect(topleft = (135,613))
1356.
1357.         cart2 = pygame.image.load('graphics\\addtocart.png').convert_alpha()
1358.         cart2_rect = cart2.get_rect(topleft = (423,613))
1359.
1360.         cart3 = pygame.image.load('graphics\\addtocart.png').convert_alpha()
1361.         cart3_rect = cart3.get_rect(topleft = (720,613))
1362.
1363.         home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
1364.         home_rect = home.get_rect(topleft = (17,8))
1365.
1366.         homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
1367.         homehov_rect = homehov.get_rect(topleft = (11,8))
1368.
1369.         cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
1370.         cartmain_rect = cartmain.get_rect(topleft = (882,8))
1371.
1372.         username_disp = sub_font.render(username,True, "White")
1373.         username_rect = username_disp.get_rect(topright = (872,7))
1374.
1375.         cartnumber = number_font.render(str(cartno),True,"White")
1376.         cartnumber_rect = cartnumber.get_rect(center = (902,26))
1377.
1378.         while True:
1379.
1380.             screen.blit(background,(0,0))
1381.             screen.blit(home,home_rect)
1382.             screen.blit(cart1,card1_rect)
1383.             screen.blit(cart2,card2_rect)
1384.             screen.blit(cart3,card3_rect)
1385.             screen.blit(cartmain,cardmain_rect)
1386.             screen.blit(username_disp,username_rect)
1387.             screen.blit(cartnumber,cardnumber_rect)
1388.
1389.             for event in pygame.event.get():
1390.                 if event.type == pygame.QUIT:
1391.                     pygame.quit()
1392.                     sys.exit()
1393.                 if event.type ==pygame.MOUSEBUTTONDOWN:
1394.                     if home_rect.collidepoint(event.pos):

```

```

1395.         shop_main()
1396.     if event.type ==pygame.MOUSEMOTION:
1397.         if home_rect.collidepoint(event.pos):
1398.             screen.blit(homehov,homehov_rect)
1399.     if event.type ==pygame.MOUSEBUTTONDOWN:
1400.         if cart1_rect.collidepoint(event.pos):
1401.             cartno+=1
1402.             cartlist[201]+=1
1403.     if event.type ==pygame.MOUSEBUTTONDOWN:
1404.         if cart2_rect.collidepoint(event.pos):
1405.             cartno+=1
1406.             cartlist[202]+=1
1407.     if event.type ==pygame.MOUSEBUTTONDOWN:
1408.         if cart3_rect.collidepoint(event.pos):
1409.             cartno+=1
1410.             cartlist[203]+=1
1411.     if event.type ==pygame.MOUSEBUTTONDOWN:
1412.         if cartmain_rect.collidepoint(event.pos):
1413.             checkout_bill()
1414.
1415.
1416.
1417.     pygame.display.update()
1418.     clock.tick(60)
1419.
1420. def warning():
1421.     global username, cartlist, cartno, merchandise,scales,scales_rect
1422.     pygame.display.set_caption("MAMBA ARCADE")
1423.     pygame.display.set_icon(ICON)
1424.     background = pygame.image.load('graphics\\warning.png').convert_alpha()
1425.     a = 0
1426.     while True:
1427.         a+=2
1428.         for event in pygame.event.get():
1429.             if event.type == pygame.QUIT:
1430.                 pygame.quit()
1431.                 sys.exit()
1432.             if a==10:
1433.                 startup()
1434.
1435.         screen.blit(background,(0,0))
1436.
1437.         pygame.display.update()
1438.         clock.tick(1)
1439.
1440. def startup():
1441.     global username, cartlist, cartno, merchandise,scales,scales_rect
1442.     pygame.display.set_caption("MAMBA ARCADE")
1443.     pygame.display.set_icon(ICON)
1444.     background = pygame.image.load('graphics\\startup.png').convert_alpha()
1445.
1446.
1447.     while True:
1448.         for event in pygame.event.get():
1449.             if event.type == pygame.QUIT:
1450.                 pygame.quit()
1451.                 sys.exit()
1452.             if event.type ==pygame.MOUSEBUTTONDOWN:
1453.                 mamba_login()
1454.             if event.type ==pygame.KEYDOWN:
1455.                 mamba_login()
1456.
1457.         screen.blit(background,(0,0))
1458.
1459.         pygame.display.update()
1460.         clock.tick(60)
1461.
1462.

```

```

1463. def shop_main():
1464.     global username, cartlist, cartno, merchandise
1465.     pygame.display.set_icon(ICON)
1466.     pygame.display.set_caption("MAMBA SHOP")
1467.     background = pygame.image.load('graphics\\shopmain.png').convert_alpha()
1468.
1469.     shirt = pygame.image.load('graphics\\shirt.png').convert_alpha()
1470.     shirt_rect = shirt.get_rect(topleft = (112,263))
1471.
1472.     mugs = pygame.image.load('graphics\\mugs.png').convert_alpha()
1473.     mugs_rect = mugs.get_rect(topleft = (384,263))
1474.
1475.     energydrinks =
pygame.image.load('graphics\\energydrinks.png').convert_alpha()
1476.     energydrinks_rect = energydrinks.get_rect(topleft = (662,263))
1477.
1478.     home = pygame.image.load('graphics\\homebutton.png').convert_alpha()
1479.     home_rect = home.get_rect(topleft = (17,8))
1480.
1481.     homehov = pygame.image.load('graphics\\homehov.png').convert_alpha()
1482.     homehov_rect = home.get_rect(topleft = (11,8))
1483.
1484.     cartmain = pygame.image.load('graphics\\cart.png').convert_alpha()
1485.     cartmain_rect = cartmain.get_rect(topleft = (882,8))
1486.
1487.     username_disp = sub_font.render(username,True, "White")
1488.     username_rect = username_disp.get_rect(topright = (872,7))
1489.
1490.     cartnumber = number_font.render(str(cartno),True,"White")
1491.     cartnumber_rect = cartnumber.get_rect(center = (902,26))
1492.
1493.
1494.     while True:
1495.
1496.         screen.blit(background,(0,0))
1497.         screen.blit(home,home_rect)
1498.         screen.blit(shirt,shirt_rect)
1499.         screen.blit(mugs,mugs_rect)
1500.         screen.blit(energydrinks,energydrinks_rect)
1501.         screen.blit(cartmain,cartmain_rect)
1502.         screen.blit(username_disp,username_rect)
1503.         screen.blit(cartnumber,cartnumber_rect)
1504.
1505.
1506.         for event in pygame.event.get():
1507.             if event.type == pygame.QUIT:
1508.                 pygame.quit()
1509.                 sys.exit()
1510.             if event.type ==pygame.MOUSEBUTTONDOWN:
1511.                 if home_rect.collidepoint(event.pos):
1512.                     mamba_main()
1513.             if event.type ==pygame.MOUSEBUTTONDOWN:
1514.                 if shirt_rect.collidepoint(event.pos):
1515.                     shop_shirt()
1516.             if event.type ==pygame.MOUSEBUTTONDOWN:
1517.                 if mugs_rect.collidepoint(event.pos):
1518.                     shop_mugs()
1519.             if event.type ==pygame.MOUSEBUTTONDOWN:
1520.                 if energydrinks_rect.collidepoint(event.pos):
1521.                     shop_energy()
1522.             if event.type ==pygame.MOUSEMOTION:
1523.                 if home_rect.collidepoint(event.pos):
1524.                     screen.blit(homehov,homehov_rect)
1525.             if event.type ==pygame.MOUSEBUTTONDOWN:
1526.                 if cartmain_rect.collidepoint(event.pos):
1527.                     checkout_bill()
1528.             mysql.commit()
1529.

```

```

1530.
1531.
1532.     pygame.display.update()
1533.     clock.tick(60)
1534.
1535. def mambarunner():
1536.     global username
1537.     pygame.display.set_icon(ICON)
1538.     screen = pygame.display.set_mode((1000, 700))
1539.     pygame.display.set_caption('MAMBARUN')
1540.     score_font = pygame.font.Font('fonts\\adrip1.ttf',73)
1541.
1542.     run1 = pygame.image.load('graphics/run1.png').convert_alpha()
1543.     run2 = pygame.image.load('graphics/run2.png').convert_alpha()
1544.     jump_image = pygame.image.load('graphics/jump.png').convert_alpha()
1545.
1546.     back = pygame.image.load('graphics/backgames.png').convert_alpha()
1547.
1548.     playagainrunner =
pygame.image.load('graphics/playagain.png').convert_alpha()
1549.     savescore = pygame.image.load('graphics/savescore.png').convert_alpha()
1550.     gamescore = pygame.image.load('graphics/score.png').convert_alpha()
1551.     playagainrunner_rect = playagainrunner.get_rect(topleft = (365,395))
1552.     gamescore_rect = gamescore.get_rect(topleft = (285,230))
1553.     savescore_rect = savescore.get_rect(topleft = (418,152))
1554.     back_rect = back.get_rect(topleft = (428,499))
1555.
1556.     runnerendscreen =
pygame.image.load('graphics/runnerendscreen.png').convert_alpha()
1557.     runnerendscreen_rect = runnerendscreen.get_rect(topleft = (0,0))
1558.
1559.     player_images = [run1, run2]
1560.     player_index = 0
1561.     player_image = player_images[player_index]
1562.     player_rect = player_image.get_rect(midbottom=(150, 580))
1563.     player_y_velocity = 0
1564.     is_jumping = False
1565.
1566.     bird = pygame.image.load('graphics/bird.png').convert_alpha()
1567.
1568.     spike= pygame.image.load('graphics/spike.png').convert_alpha()
1569.
1570.     background_x = 0
1571.     background = pygame.image.load('graphics/download.jpg').convert_alpha()
1572.     background_rect = background.get_rect(topleft = (background_x,0))
1573.
1574.
1575.     clock = pygame.time.Clock()
1576.     gravity = 1
1577.     jump_strength = -30
1578.     obstacle_speed = 7
1579.     obstacle_timer = pygame.USEREVENT + 1
1580.     pygame.time.set_timer(obstacle_timer, 1500)
1581.     obstacles = []
1582.     global score
1583.     score = scoretime = 0
1584.     font = pygame.font.Font(None, 36)
1585.
1586.     def endgame():
1587.         global score
1588.         scored = str(score)
1589.         scoredisp = score_font.render(scored,True,'White')
1590.
1591.         while True:
1592.             for event in pygame.event.get():
1593.                 if event.type == pygame.QUIT:
1594.                     pygame.quit()
1595.                     sys.exit()

```

```

1596.
1597.         if event.type ==pygame.MOUSEBUTTONDOWN:
1598.             if back_rect.collidepoint(event.pos):
1599.                 games_main()
1600.         if event.type ==pygame.MOUSEBUTTONDOWN:
1601.             if playagainrunner_rect.collidepoint(event.pos):
1602.                 mambarunner()
1603.         if event.type ==pygame.MOUSEBUTTONDOWN:
1604.             if savescore_rect.collidepoint(event.pos):
1605.                 cursor.execute("UPDATE SCORES SET GEORGERUNS = %s WHERE
USERNAME = %s", (score, username))
1606.                 mysql.commit()
1607.
1608.
1609.
1610.             screen.blit(runnerendscreen,runnerendscreen_rect)
1611.             screen.blit(playagainrunner,playagainrunner_rect)
1612.             screen.blit(back,back_rect)
1613.             screen.blit(gamescore,gamescore_rect)
1614.             screen.blit(scoredisp,(505,230))
1615.             if username != 'guest':
1616.                 screen.blit(savescore,savescore_rect)
1617.
1618.
1619.
1620.             pygame.display.update()
1621.             clock.tick(60)
1622.
1623.         def create_obstacle():
1624.             if ran.choice(['bird', 'spike']) == 'bird':
1625.                 obstacle_rect = bird.get_rect(midbottom=(ran.randint(1100, 1300),
ran.randint(400, 550)))
1626.             else:
1627.                 obstacle_rect = spike.get_rect(midbottom=(ran.randint(1100, 1300),
580))
1628.                 obstacles.append(obstacle_rect)
1629.
1630.         def move_obstacles(obstacles):
1631.             for obstacle in obstacles:
1632.                 obstacle.x -= obstacle_speed
1633.                 obstacles = [obstacle for obstacle in obstacles if obstacle.x > -50]
1634.             return obstacles
1635.
1636.         def draw_obstacles(obstacles):
1637.             for obstacle in obstacles:
1638.                 if obstacle.bottom < 580:
1639.                     screen.blit(bird, obstacle)
1640.                 else:
1641.                     screen.blit(spike, obstacle)
1642.
1643.         def check_collision(obstacles):
1644.             for obstacle in obstacles:
1645.                 if player_rect.colliderect(obstacle):
1646.                     return False
1647.             return True
1648.
1649.         while True:
1650.             for event in pygame.event.get():
1651.                 if event.type == pygame.QUIT:
1652.                     pygame.quit()
1653.                     sys.exit()
1654.
1655.                 if event.type == obstacle_timer:
1656.                     create_obstacle()
1657.
1658.             keys = pygame.key.get_pressed()
1659.             if keys[pygame.K_SPACE] and player_rect.bottom >= 580:
1660.                 is_jumping = True

```

```

1661.         player_y_velocity = jump_strength
1662.
1663.     if is_jumping:
1664.         player_y_velocity += gravity
1665.         player_rect.y += player_y_velocity
1666.         if player_rect.bottom >= 580:
1667.             player_rect.bottom = 580
1668.             is_jumping = False
1669.
1670.     if not is_jumping:
1671.         player_index += 0.1
1672.         if player_index >= len(player_images):
1673.             player_index = 0
1674.         player_image = player_images[int(player_index)]
1675.     else:
1676.         player_image = jump_image
1677.
1678.     obstacles = move_obstacles(obstacles)
1679.     screen.blit(background,background_rect)
1680.     draw_obstacles(obstacles)
1681.
1682.     game_active = check_collision(obstacles)
1683.     if not game_active:
1684.         endgame()
1685.
1686.     background_x+/-10
1687.
1688.
1689.     screen.blit(player_image, player_rect)
1690.
1691.     score_text = font.render(f'Score: {score}', True, (255, 255, 255))
1692.     screen.blit(score_text, (850, 10))
1693.     scoretime += 0.16
1694.     score = int(scoretime)
1695.
1696.     pygame.display.update()
1697.     clock.tick(60)
1698.
1699.
1700. def poppeye_game():
1701.     def poppeye_end():
1702.         score_font = pygame.font.Font('fonts\\adripl.ttf',73)
1703.         back = pygame.image.load('graphics/backgames.png').convert_alpha()
1704.         playagain = pygame.image.load('graphics/playagain.png').convert_alpha()
1705.         savescore = pygame.image.load('graphics/savescore.png').convert_alpha()
1706.         gamescore = pygame.image.load('graphics/score.png').convert_alpha()
1707.         savescore = pygame.image.load('graphics/savescore.png').convert_alpha()
1708.         playagain_rect = playagain.get_rect(topleft = (365,395))
1709.         gamescore_rect = gamescore.get_rect(topleft = (285,230))
1710.         savescore_rect = savescore.get_rect(topleft = (418,152))
1711.         back_rect = back.get_rect(topleft = (428,499))
1712.         poppeye_end =
pygame.image.load('graphics/poppeye_end.png').convert_alpha()
1713.         poppeye_end_rect = poppeye_end.get_rect(topleft = (0,0))
1714.
1715.         scored = str(score)
1716.         scoredisp = score_font.render(scored,True,'White')
1717.
1718.         while True:
1719.             for event in pygame.event.get():
1720.                 if event.type == pygame.QUIT:
1721.                     pygame.quit()
1722.                     sys.exit()
1723.
1724.                 if event.type ==pygame.MOUSEBUTTONDOWN:
1725.                     if back_rect.collidepoint(event.pos):
1726.                         games_main()
1727.                 if event.type ==pygame.MOUSEBUTTONDOWN:

```

```

1728.             if playagain_rect.collidepoint(event.pos):
1729.                 poppeye_game()
1730.             if event.type ==pygame.MOUSEBUTTONDOWN:
1731.                 if savescore_rect.collidepoint(event.pos):
1732.                     cursor.execute("UPDATE SCORES SET POPPEYE = %s WHERE
USERNAME = %s", (score, username))
1733.                     mysql.commit()
1734.
1735.
1736.
1737.             screen.blit(poppeye_end,poppeye_end_rect)
1738.             screen.blit(playagain,playagain_rect)
1739.             screen.blit(back,back_rect)
1740.             screen.blit(gamescore,gamescore_rect)
1741.             screen.blit(scoredisp,(505,230))
1742.             if username != 'guest':
1743.                 screen.blit(savescore,savescore_rect)
1744.
1745.
1746.             pygame.display.update()
1747.             clock.tick(60)
1748.
1749.             screen = pygame.display.set_mode((1000, 700))
1750.             pygame.display.set_caption('2D Game')
1751.
1752.             background = pygame.image.load('graphics/background.jpg')
1753.
1754.             char = pygame.image.load('graphics/popeye.png').convert_alpha()
1755.             player = pygame.transform.scale(char, (170, 272))
1756.             player_rect = player.get_rect(midbottom=(500, 650))
1757.
1758.             char_l = pygame.image.load('graphics/popeye_left.png').convert_alpha()
1759.             player_l = pygame.transform.scale(char_l, (300, 272))
1760.             player_l_rect = player.get_rect(bottomright=(500, 650))
1761.
1762.             char_r = pygame.image.load('graphics/popeye_right.png').convert_alpha()
1763.             player_r = pygame.transform.scale(char_r, (300, 272))
1764.             player_r_rect = player.get_rect(midbottom=(500, 650))
1765.
1766.             oli = pygame.image.load('graphics/olive.png').convert_alpha()
1767.             olive = pygame.transform.scale(oli, (140, 211))
1768.
1769.             steak_image = pygame.image.load('graphics/steak.png').convert_alpha()
1770.             steak = pygame.transform.scale(steak_image, (70, 70))
1771.             sandwich_image = pygame.image.load('graphics/sandwich.png').convert_alpha()
1772.             sandwich = pygame.transform.scale(sandwich_image, (70, 70))
1773.             wine_image = pygame.image.load('graphics/wine.png').convert_alpha()
1774.             wine = pygame.transform.scale(wine_image, (100, 100))
1775.             hotdog_image = pygame.image.load('graphics/hotdog.png').convert_alpha()
1776.             hotdog = pygame.transform.scale(hotdog_image, (70, 70))
1777.
1778.             oli_throw = pygame.image.load('graphics/olive_throw.png').convert_alpha()
1779.             olive_throw = pygame.transform.scale(oli_throw, (140, 211))
1780.
1781.             goat = pygame.image.load('graphics/boat.png').convert_alpha()
1782.             boat = pygame.transform.scale(goat, (400, 400))
1783.             boat_rect = boat.get_rect(midbottom=(500, 700))
1784.
1785.             plat = pygame.image.load('graphics/platform.png').convert_alpha()
1786.             platform = pygame.transform.scale(plat, (165, 50))
1787.
1788.             image = pygame.transform.scale(background, (1000, 700))
1789.
1790.             clock = pygame.time.Clock()
1791.             pos = 0
1792.
1793.             olive_switch_timer = pygame.time.get_ticks()
1794.             olive_display_throw = False

```

```

1795.     switch_interval = 700
1796.
1797.     items = []
1798.
1799.     gravity = 0.175
1800.     item_speed = 8
1801.
1802.     score = 0
1803.     lives = 3
1804.
1805.     font = pygame.font.Font(None, 36)
1806.
1807.     target_positions = [500,1000,2400]
1808.
1809.     catch_rects = {-1: pygame.Rect(300, 550, 100, 100),0: pygame.Rect(450, 550,
1810. 100, 100),1: pygame.Rect(650, 550, 100, 100)}
1811.
1812.     while True:
1813.         for event in pygame.event.get():
1814.             if event.type == pygame.QUIT:
1815.                 pygame.quit()
1816.                 sys.exit()
1817.
1818.             if event.type == pygame.KEYDOWN:
1819.                 if event.key == pygame.K_a:
1820.                     if pos > -1:
1821.                         pos -= 1
1822.                 elif event.key == pygame.K_d:
1823.                     if pos < 1:
1824.                         pos += 1
1825.
1826.         current_time = pygame.time.get_ticks()
1827.         if current_time - olive_switch_timer > switch_interval:
1828.             olive_display_throw = not olive_display_throw
1829.             olive_switch_timer = current_time
1830.
1831.             if olive_display_throw:
1832.                 item_type = ran.choice(['steak', 'sandwich', 'wine', 'hotdog'])
1833.                 if item_type == "steak":
1834.                     item_image = steak
1835.                 elif item_type == "sandwich":
1836.                     item_image = sandwich
1837.                 elif item_type == "hotdog":
1838.                     item_image = hotdog
1839.                 else:
1840.                     item_image = wine
1841.
1842.                 target_x = ran.choice(target_positions)
1843.                 target_y = 800
1844.
1845.                 direction_x = target_x - 80
1846.                 direction_y = target_y - 100
1847.
1848.                 distance = math.sqrt(direction_x**2 + direction_y**2)
1849.                 direction_x /= distance
1850.                 direction_y /= distance
1851.
1852.                 velocity_x = direction_x * item_speed
1853.                 velocity_y = direction_y * item_speed - 4
1854.
1855.                 items.append({"image": item_image, "pos": [80, 100],
1856. "velocity": [velocity_x, velocity_y], "target": target_x})
1857.
1858.                 screen.blit(image, (0, 0))
1859.
1860.             if olive_display_throw:
1861.                 screen.blit(olive_throw, (60, 70))
1862.             else:

```

```

1861.         screen.blit(olive, (0, 70))
1862.
1863.         if pos == 0:
1864.             screen.blit(player, player_rect)
1865.         elif pos == 1:
1866.             screen.blit(player_r, player_r_rect)
1867.         elif pos == -1:
1868.             screen.blit(player_l, player_l_rect)
1869.
1870.         screen.blit(boat, boat_rect)
1871.         screen.blit(platform, (-20, 260))
1872.
1873.
1874.         player_catch_rect = catch_rects[pos]
1875.
1876.         for i in items[:]:
1877.             i["pos"][0] += i["velocity"][0]
1878.             i["pos"][1] += i["velocity"][1]
1879.             i["velocity"][1] += gravity
1880.
1881.             screen.blit(i["image"], i["pos"])
1882.
1883.             if player_catch_rect.colliderect(pygame.Rect(i["pos"][0],
i["pos"][1], i["image"].get_width(), i["image"].get_height())):
1884.                 score += 1
1885.                 items.remove(i)
1886.
1887.             elif i["pos"][1] > 700:
1888.                 lives -= 1
1889.                 items.remove(i)
1890.
1891.         score_text = font.render(f'Score: {score}', True, (255, 255, 255))
1892.         lives_text = font.render(f'Lives: {lives}', True, (255, 255, 255))
1893.         screen.blit(score_text, (10, 10))
1894.         screen.blit(lives_text, (10, 50))
1895.
1896.         if lives <= 0:
1897.             poppeye_end()
1898.
1899.
1900.         pygame.display.update()
1901.         clock.tick(60)
1902.
1903.
1904.
1905.     def mambaio_game():
1906.         mambaio_background =
pygame.image.load('graphics/mambaio_background.png').convert_alpha()
1907.         mambaio_background_rect = mambaio_background.get_rect(topleft = (0,0))
1908.
1909.         width, height = 1000, 700
1910.         block_size, border_thickness = 20, 5
1911.         colors = {
1912.             'snake_body': (0, 0, 0),
1913.             'snake_stripe': (255, 255, 255),
1914.             'food': (255, 255, 0),
1915.             'straw': (255, 0, 0),
1916.             'bg': (128, 128, 128),
1917.             'border': (0, 0, 0)
1918.         }
1919.         directions = {'up': (0, -1), 'down': (0, 1), 'left': (-1, 0), 'right': (1,
0)}
1920.
1921.         pygame.init()
1922.         screen = pygame.display.set_mode((width, height))
1923.         pygame.display.set_caption("Snake Game")
1924.
1925.         snake = [(100, 100), (80, 100), (60, 100)]

```

```

1926.     direction, new_direction = directions['right'], directions['right']
1927.     score = 0
1928.     score_font = pygame.font.Font('fonts\\adripl.TTF',43)
1929.     food_pos = (
1930.         ran.randint(0, (width // block_size) - 1) * block_size,
1931.         ran.randint(0, (height // block_size) - 1) * block_size
1932.     )
1933.
1934.     def draw():
1935.         screen.blit(mambaio_background,mambaio_background_rect)
1936.         for i, (x, y) in enumerate(snake):
1937.             color = colors['snake_body'] if i % 2 == 0 else
colors['snake_stripe']
1938.             pygame.draw.rect(screen, color, (x, y, block_size, block_size))
1939.             pygame.draw.rect(screen, colors['border'], (x, y, block_size,
block_size), 1)
1940.             head_x, head_y = snake[0]
1941.             eye_r, eye_offset_x, eye_offset_y = block_size // 6, block_size // 3,
block_size // 4
1942.             pygame.draw.circle(screen, (255, 255, 255), (head_x + eye_offset_x,
head_y + eye_offset_y), eye_r)
1943.             pygame.draw.circle(screen, (255, 255, 255), (head_x + 2 * eye_offset_x,
head_y + eye_offset_y), eye_r)
1944.             food_x, food_y = food_pos
1945.             pygame.draw.rect(screen, colors['food'], (food_x, food_y, block_size,
block_size))
1946.             straw_x, straw_y = food_x + block_size // 2, food_y
1947.             pygame.draw.line(screen, colors['straw'], (straw_x, straw_y), (straw_x,
straw_y - 10), 3)
1948.             score_display = score_font.render(f"Score: {score}", True,
(255,255,255))
1949.             screen.blit(score_display, (width - 150, 10))
1950.
1951.     def flash():
1952.         screen.fill((255, 255, 255))
1953.         pygame.display.flip()
1954.         pygame.time.delay(100)
1955.         screen.fill(colors['bg'])
1956.
1957.     def game_over():
1958.         score_font = pygame.font.Font('fonts\\adripl.ttf',73)
1959.         back = pygame.image.load('graphics/backgames.png').convert_alpha()
1960.         playagain = pygame.image.load('graphics/playagain.png').convert_alpha()
1961.         savescore = pygame.image.load('graphics/savescore.png').convert_alpha()
1962.         gamescore = pygame.image.load('graphics/score.png').convert_alpha()
1963.         savescore = pygame.image.load('graphics/savescore.png').convert_alpha()
1964.         playagain_rect = playagain.get_rect(topleft = (365,395))
1965.         gamescore_rect = gamescore.get_rect(topleft = (285,230))
1966.         savescore_rect = savescore.get_rect(topleft = (418,152))
1967.         back_rect = back.get_rect(topleft = (428,499))
1968.         mambaio_end =
pygame.image.load('graphics/mambaio_end.png').convert_alpha()
1969.         mambaio_end_rect = mambaio_end.get_rect(topleft = (0,0))
1970.
1971.         scored = str(score)
1972.         scoredisp = score_font.render(scored,True,'White')
1973.         while True:
1974.             for event in pygame.event.get():
1975.                 if event.type == pygame.QUIT:
1976.                     pygame.quit()
1977.                     sys.exit()
1978.
1979.                 if event.type ==pygame.MOUSEBUTTONDOWN:
1980.                     if back_rect.collidepoint(event.pos):
1981.                         games_main()
1982.                 if event.type ==pygame.MOUSEBUTTONDOWN:
1983.                     if playagain_rect.collidepoint(event.pos):
1984.                         mambaio_game()

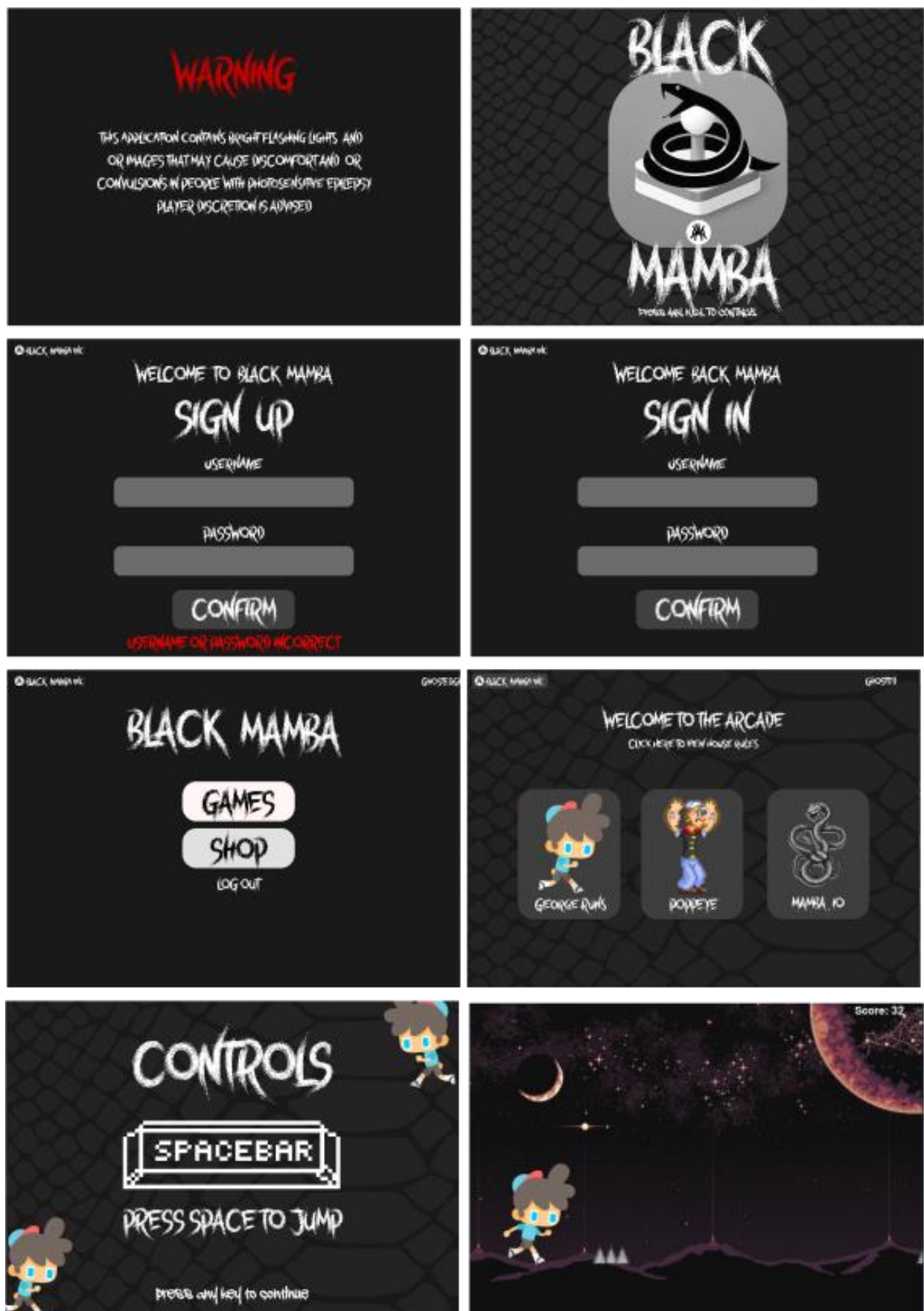
```

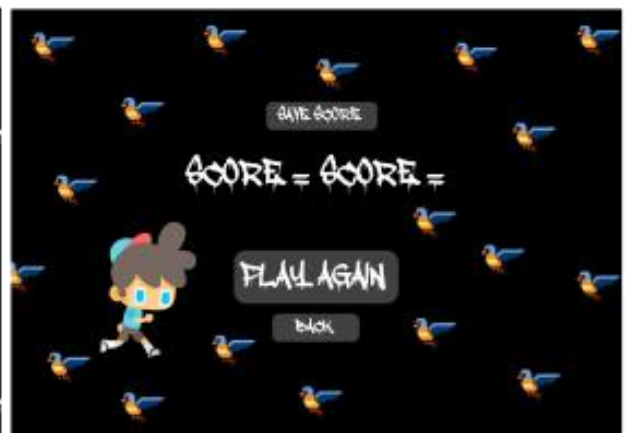
```

1985.         if event.type ==pygame.MOUSEBUTTONDOWN:
1986.             if savescore_rect.collidepoint(event.pos):
1987.                 cursor.execute("UPDATE SCORES SET MAMBAIO = %s WHERE
USERNAME = %s", (score, username))
1988.                 mysql.commit()
1989.
1990.
1991.
1992.         screen.blit(mambaio_end,mambaio_end_rect)
1993.         screen.blit(playagain,playagain_rect)
1994.         screen.blit(back,back_rect)
1995.         screen.blit(gamescore,gamescore_rect)
1996.         screen.blit(scoredisp,(505,230))
1997.         if username != 'guest':
1998.             screen.blit(savescore,savescore_rect)
1999.
2000.
2001.
2002.         pygame.display.update()
2003.         clock.tick(60)
2004.
2005.     speed = 1
2006.     while True:
2007.
2008.         for event in pygame.event.get():
2009.             if event.type == pygame.QUIT:
2010.                 pygame.quit()
2011.                 sys.exit()
2012.             elif event.type == pygame.KEYDOWN:
2013.                 if event.key == pygame.K_w and direction != directions['down']:
2014.                     new_direction = directions['up']
2015.                 elif event.key == pygame.K_s and direction != directions['up']:
2016.                     new_direction = directions['down']
2017.                 elif event.key == pygame.K_a and direction !=
directions['right']:
2018.                     new_direction = directions['left']
2019.                 elif event.key == pygame.K_d and direction !=
directions['left']:
2020.                     new_direction = directions['right']
2021.
2022.
2023.                 if direction != new_direction:
2024.                     direction = new_direction
2025.                 head_x, head_y = snake[0]
2026.                 dir_x, dir_y = direction
2027.                 new_head = (head_x + dir_x * block_size, head_y + dir_y * block_size)
2028.                 snake.insert(0, new_head)
2029.                 if new_head == food_pos:
2030.                     flash()
2031.                     food_pos = (
2032.                         ran.randint(0, (width // block_size) - 1) * block_size,
2033.                         ran.randint(0, (height // block_size) - 1) * block_size
2034.                     )
2035.                     score += 1
2036.                     speed+=0.01
2037.                 else:
2038.                     snake.pop()
2039.                 if (new_head[0] < 0 or new_head[0] >= width or
2040.                     new_head[1] < 0 or new_head[1] >= height or
2041.                     new_head in snake[1:]):
2042.                     game_over()
2043.                 draw()
2044.                 pygame.display.flip()
2045.                 clock.tick(10**speed)
2046.
2047.
2048.
2049.     warning()

```

Output:







MySQL Database Table Structures

Query to setup database and tables in sql:

create database blackmamba; use blackmamba; create table credentials(username varchar(15) primary key, password varchar(15), email varchar(60), address varchar(80), phoneno varchar(80), ccno varchar(19), expirydate varchar(6), cvv varchar(4)); create table scores(username varchar(15) primary key, georgeruns int, poppeye int, mambaio int);

```
mysql> use blackmamba;
Database changed
mysql> show tables;
+-----+
| Tables_in_blackmamba |
+-----+
| credentials           |
| scores                 |
+-----+
2 rows in set (0.04 sec)

mysql> desc credentials;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| username   | varchar(15)   | NO   | PRI | NULL    |       |
| password   | varchar(15)   | YES  |     | NULL    |       |
| email      | varchar(60)   | YES  |     | NULL    |       |
| address    | varchar(80)   | YES  |     | NULL    |       |
| phoneno    | bigint        | YES  |     | NULL    |       |
| ccno       | varchar(19)   | YES  |     | NULL    |       |
| expirydate | varchar(6)    | YES  |     | NULL    |       |
| cvv        | varchar(4)    | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.02 sec)

mysql> desc scores;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| username   | varchar(20)   | NO   | PRI | NULL    |       |
| georgeruns | int           | YES  |     | NULL    |       |
| poppeye    | int           | YES  |     | NULL    |       |
| mambaio    | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

Errors Identified and Experienced:

Throughout the development of *Black Mamba Arcade*, the team encountered and resolved several errors to enhance stability and user experience. Early on, syntax and runtime errors in Pygame initialization and MySQL queries were addressed to prevent unexpected application crashes. Database connection issues posed a challenge as well, with some functions failing due to incorrect cursor execution or misconfigured connection parameters; these were refined for reliable data retrieval and storage. User interface inconsistencies were another area of improvement, especially in ensuring button responsiveness and seamless navigation between screens. Adjustments to collision detection in games and score-tracking logic were necessary to provide accurate gameplay feedback. The team also optimized resource loading and asset management to reduce lag and improve graphics rendering, ensuring smoother transitions and overall application performance. Through careful debugging and iterative testing, these issues were systematically fixed, resulting in a more robust and enjoyable experience for users.

Future Modifications:

Looking ahead, *Black Mamba Arcade* has exciting potential for future enhancements that would enrich user engagement and security. Implementing a leaderboard and high score system would allow players to compete for top scores, fostering a sense of community and motivation. Additionally, higher quality animations could make gameplay visually captivating, creating a more immersive experience. To improve account security, introducing two-step verification and a “forgot password” feature would add layers of protection, ensuring that user data remains safe. The addition of in-game currency could provide players with a rewarding experience, enabling them to purchase virtual items or power-ups, which would elevate engagement and provide further incentives. Together, these updates could enhance both the enjoyment and security of *Black Mamba Arcade*, making it a more dynamic, user-centered application.

Credits:

The development of Black Mamba Arcade was a collaborative effort by a dedicated team. Arun Jose, serving as both the project leader and lead creative director, skillfully handled both the backend and frontend, designing everything from scratch to create a cohesive user interface that provides an intuitive and enjoyable user experience. Kavin Senthilkumar contributed to the creation of engaging arcade games, enhancing the application's entertainment value. Aditya Menon focused on implementing the game logic and establishing seamless MySQL connectivity, ensuring efficient data management and performance. Together, their combined efforts brought Black Mamba Arcade to life, creating a valuable tool for stress relief and financial education.

Bibliography:

- Ncert Class 12 Computer Science
- Computer Science with Python Class 12 Sumita Arora
- <https://www.youtube.com/watch?v=AY9MnQ4x3zk>
- <https://www.youtube.com/watch?v=U0a83bRJw6o>

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